

Sequential Randomization Tests Using e-values: Applications for trial monitoring

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April 29, 2026

Abstract

Sequential monitoring of randomized trials traditionally relies on parametric assumptions or asymptotic approximations. We discuss a family of nonparametric sequential tests—collectively called e-RT—for binary, event-only, continuous, and time-to-event endpoints. All variants derive validity from the randomization mechanism. Using a betting framework, each test constructs a test martingale by sequentially wagering on randomized assignments or randomized event identities before using the current label in the wealth update. Under the null hypothesis of no treatment effect, the expected wealth cannot grow, guaranteeing anytime-valid Type I error control regardless of stopping rule. The default e-RT posture is effect-size agnostic: monitoring can begin without specifying a hypothesized treatment effect. Alternatively, fixed design-calibrated wagers, including growth-rate-optimal (GROW) wagers, may be used as optional efficiency tools when a clinically meaningful design alternative is credible. We present simulation studies demonstrating calibration and power, and discuss the principled asymmetry in betting strategies across outcome types. These methods provide a conservative, assumption-light complement to model-based sequential analyses.

Keywords: e-values, e-process, randomization test, sequential analysis, clinical trials

1 Introduction

Sequential monitoring of randomized controlled trials requires methods that control Type I errors regardless of when or why the monitoring stops. Traditional group-sequential designs rely on parametric assumptions and predetermined stopping boundaries. When these assumptions fail, or when trials adapt in ways not fully prespecified, validity guarantees may erode.

Monitoring clinical trials, especially in the context of acutely ill patients, is of paramount importance. Traditional methods include α – *spending* functions. When interim analyses are spaced, evidence grows unnoticed between the interim analyses. This may both delay the implementation of potentially useful strategies or prolong trials when safety signals arise early.

E-values and e-processes offer an alternative framework (Shafer, 2021; Vovk and Wang, 2021; Ramdas and Wang, 2025). An e-value is a measure of evidence against a null hypothesis with a specific property: its expected value under the null is at most 1. This simple constraint yields anytime-valid inference: the Type I error guarantee holds at any stopping time, regardless of the stopping rule.

Duan et al. (2022) introduced interactive rank testing by betting (i-bet), which tests treatment effects by wagering on treatment assignments given observed outcomes. The intuition is discussed by Ramdas (2021). Under the null hypothesis, randomization ensures that assignments are independent of outcomes, so no betting strategy can systematically accumulate wealth.

Sokolova and Sokolov (2026a) recently developed a practitioner-oriented framework for e-value monitoring in adaptive clinical trials, including design-calibrated binary e-processes, safe logrank monitoring, futility tools, platform-trial extensions, and an open-source implementation in the `evalinger` package. Their work emphasizes a design-calibrated view of e-process construction: the betting strategy is chosen to optimize expected evidence

growth under a prespecified clinically meaningful alternative. This connects to growth-rate-optimal (GROW) wagering in the broader e-value literature (Ramdas and Wang, 2025), and provides the central external contrast for the present work: e-RT is effect-size agnostic by default, while design-calibrated wagers are treated here as optional efficiency tools rather than as prerequisites for monitoring.

We propose e-RT—recursively, e-RT is a Randomization Test—as a complementary family of methods for prospective sequential monitoring of randomized trials. Like i-bet (Duan et al., 2022), e-RT uses betting martingales for inference, but differs in key respects: e-RT monitors sequentially as patients enroll rather than analyzing completed trial data; it requires no covariates or working models for validity; and its default wager policies can be learned from accumulating data rather than fixed by a hypothesized effect size. This yields methods with minimal assumptions suitable for real-time trial monitoring. Design-calibrated wagers are then added as optional efficiency modes, not as the defining feature of the method.

We describe four active variants: e-RTb for binary event/no-event outcomes; e-RTe for event-only monitoring (requiring no non-event tracking); e-RTc for continuous endpoints; and e-RTs for time-to-event data. All active variants share the same validity proof—the expected wealth multiplier is exactly 1 under the null—but differ in how they translate outcome data into wagers. We also distinguish the randomization-based validity engine from the wager policy. The default adaptive policies preserve effect-size agnosticism. Prespecified design alternatives can be used to choose more aggressive wagers when those assumptions are credible, but this is an efficiency choice rather than a validity requirement. This distinction is especially relevant in sparse-update settings, where fixed or design-calibrated wagers may improve power without the same over-betting sensitivity seen with dense every-patient updates.

2 Unified validity argument

All e-RT variants in this manuscript use the same mathematical device. At each monitoring step, the analyst observes some information that is allowed to inform the wager, then a randomized label or exchangeable sign is revealed, and wealth is multiplied by a fair-payoff factor. The endpoint-specific sections differ only in what the observed signal is and how the wager is chosen.

Let $\mathcal{F}_{k-1}$ denote all information revealed after the first $k-1$ betting updates. Before update k , the analyst may observe a signal S_k that does not reveal the random label being bet on. Examples include the current binary outcome before revealing treatment assignment, the fact that a new event has occurred before revealing the event arm, a continuous outcome before revealing assignment, or a risk set before the next event identity. Let

$$\mathcal{G}_k = \sigma(\mathcal{F}_{k-1}, S_k) \tag{1}$$

be the pre-reveal information at update k . A wager is valid if it is $\mathcal{G}_k$ -measurable and the resulting multiplier is nonnegative. In this manuscript, we call such wagers *predictable*: they may depend on prior data, current pre-reveal signals, randomization probabilities, risk sets, and prespecified design alternatives, but not on the label or sign that determines whether the wager wins.

This reveal order is a mathematical bookkeeping device. In an open-label trial, treatment assignments may be known operationally before outcomes are observed. The requirement is not that the clinical team be blinded; it is that the algorithm used to choose the current wager must be computable without using the current label or sign being tested. Past assignments may be used by adaptive wager policies after their corresponding wealth updates have already occurred.

There are two equivalent forms used below. In assignment-prediction variants, the hidden label is $A_k \in \{0, 1\}$ with known null probability $\pi_k = P_0(A_k = 1 \mid \mathcal{G}_k)$. A betting fraction $\lambda_k \in [0, 1]$ gives multiplier

$$M_k = \begin{cases} \lambda_k / \pi_k, & A_k = 1, \\ (1 - \lambda_k) / (1 - \pi_k), & A_k = 0. \end{cases} \tag{2}$$

In score-increment variants, the endpoint produces a bounded or otherwise controlled increment U_k satisfying

$$\mathbb{E}_0(U_k \mid \mathcal{G}_k) = 0, \tag{3}$$

1 and the wager b_k is chosen so that

$$M_k = 1 + b_k U_k \geq 0. \quad (4)$$

2 **Theorem 1** (Unified e-RT validity). *If each wager is predictable in the sense above and each multiplier is*
 3 *nonnegative, then the wealth process*

$$W_k = \prod_{\ell=1}^k M_\ell, \quad W_0 = 1, \quad (5)$$

4 *is a nonnegative test martingale under the null hypothesis. Consequently, for any stopping time τ and any*
 5 *$\alpha \in (0, 1)$,*

$$P_0 \left(\sup_{k \geq 1} W_k \geq \frac{1}{\alpha} \right) \leq \alpha. \quad (6)$$

6 *Proof.* For the assignment-prediction multiplier in Equation 2,

$$\mathbb{E}_0(M_k \mid \mathcal{G}_k) = \pi_k \frac{\lambda_k}{\pi_k} + (1 - \pi_k) \frac{1 - \lambda_k}{1 - \pi_k} \quad (7)$$

$$= \lambda_k + (1 - \lambda_k) = 1. \quad (8)$$

7 For the score-increment multiplier in Equation 4,

$$\mathbb{E}_0(M_k \mid \mathcal{G}_k) = 1 + b_k \mathbb{E}_0(U_k \mid \mathcal{G}_k) = 1. \quad (9)$$

8 In either case,

$$\mathbb{E}_0(W_k \mid \mathcal{G}_k) = W_{k-1} \mathbb{E}_0(M_k \mid \mathcal{G}_k) = W_{k-1}. \quad (10)$$

9 Taking conditional expectation again with respect to $\mathcal{F}_{k-1}$ gives $\mathbb{E}_0(W_k \mid \mathcal{F}_{k-1}) = W_{k-1}$, so (W_k) is a martingale.
 10 Nonnegativity follows from the multiplier constraint. Ville's inequality for nonnegative martingales (Ville, 1939)
 11 then gives the anytime-valid Type I error bound. $\square$

12 This theorem is deliberately agnostic about how the wager is selected. An adaptive wager is valid if it is
 13 computed from past data and current pre-reveal information. A fixed or design-calibrated wager is valid if it
 14 is prespecified from protocol quantities or a design alternative before the trial begins, and then applied without
 15 looking at the current hidden label. Misspecifying the design alternative can reduce power, delay crossing, or
 16 inflate the apparent effect among crossing trials, but it does not change the conditional expectation calculation
 17 above. Validity is a property of randomization, exchangeability, predictability, and nonnegative fair multipliers;
 18 power is a property of how well the wager matches the alternative.

19 3 e-RT binary (e-RTb)

20 Binary endpoints are common in clinical trials, with mortality, clinical deterioration, infection, treatment failure,
 21 or response status often recorded as event/no-event outcomes. The binary e-RT, abbreviated **e-RTb**, is the
 22 simplest member of the family and serves as the prototype for the later variants.

23 3.1 Setup

24 Consider a sequential randomized trial with 1:1 allocation. At each enrollment $i = 1, 2, \dots$, we observe:

- 25 • $T_i \in \{0, 1\}$: treatment assignment (0 = control, 1 = intervention)
- 26 • $Y_i \in \{0, 1\}$: binary outcome (0 = no event, 1 = event)

27 Treatment is assigned with known probability $p = P(T_i = 1)$, typically $p = 0.5$.

28 The null hypothesis is that treatment assignment has no effect on outcome:

$$H_0 : Y_i \perp T_i \text{ for all } i \quad (11)$$

Under this hypothesis, observing the outcome provides no information about which arm the patient was assigned to.

3.2 Wealth Process

Following Duan et al. (2022), we construct a wealth process by wagering on treatment assignments. After observing outcome Y_i but *before* learning treatment assignment T_i , we choose $\lambda_i \in [0, 1]$: the fraction wagered on intervention.

The wealth updates as:

$$W_i = W_{i-1} \times \begin{cases} \lambda_i/p & \text{if } T_i = 1 \\ (1 - \lambda_i)/(1 - p) & \text{if } T_i = 0 \end{cases} \quad (12)$$

starting from $W_0 = 1$. When we bet toward the correct arm, wealth grows; when wrong, it shrinks.

3.3 Betting Strategy

The validity guarantee follows from the unified argument above. The wager must be chosen before the treatment assignment for patient i is used in the wealth update. It may depend on the current outcome Y_i , because the betting game is framed as predicting assignment from outcome, and it may depend on prior patients' outcomes and assignments. It may not depend on T_i itself. Power depends on choosing bets that grow wealth under the alternative. We use a strategy that learns the treatment effect from accumulating data.

Let:

$$\hat{\delta}_{i-1} = (\text{event rate in intervention}) - (\text{event rate in control}) \quad (13)$$

estimated from patients $1, \dots, i-1$. The betting fraction is:

$$\lambda_i = \begin{cases} 0.5 + 0.5 \cdot c_i \cdot \hat{\delta}_{i-1} & \text{if } Y_i = 1 \\ 0.5 - 0.5 \cdot c_i \cdot \hat{\delta}_{i-1} & \text{if } Y_i = 0 \end{cases} \quad (14)$$

where $c_i \in [0, 1]$ ramps from 0 to 1 over a burn-in period:

$$c_i = \min \left(1, \max \left(0, \frac{i - n_0}{n_r} \right) \right) \quad (15)$$

with n_0 the burn-in period and n_r the ramp period. This prevents large bets when $\hat{\delta}_{i-1}$ is unstable due to small samples.

The logic: if $\hat{\delta} > 0$ (more events in intervention), then events suggest intervention and non-events suggest control. If $\hat{\delta} < 0$ (fewer events in intervention), then events suggest control and non-events suggest intervention. The factor of 0.5 before $c_i \cdot \hat{\delta}_{i-1}$ ensures $\lambda_i \in [0, 1]$.

3.4 Worked Example

Consider a trial comparing intervention versus control, with a binary outcome (event or no event). Event is mortality, which is expected to be lower with intervention. Allocation is 1:1 ($p = 0.5$). Assume burn-in is complete ($c_i = 1$).

We look back at patients 1–199. Intervention arm has 100 patients, 35 events, so rate = 35.0%. Control arm has 99 patients, 40 events, so rate = 40.4%. $\hat{\delta}_{199} = 0.350 - 0.404 = -0.054$ (intervention looks protective).

Patient 200 has an event (dies). Where is this patient likely from? Events are more common in control (40.4% vs 35.0%), so probably control. We bet $\lambda = 0.5 + 0.5 \times (-0.054) = 0.473$ on intervention and $1 - \lambda = 0.527$ on control. Assignment revealed: control. We guessed right. Multiplier: $0.527/0.5 = 1.054$. Wealth grows 5.4%.

Patient 201: Update counts—intervention has 100 patients, 35 events (35.0%); control has 100 patients, 41 events (41.0%). $\hat{\delta}_{200} = -0.060$. Patient 201 has no event. Non-events are more common in intervention (65.0% vs 59.0%), so probably intervention. Bet: $\lambda = 0.5 - 0.5 \times (-0.060) = 0.530$. Assignment revealed: intervention. Multiplier: $0.530/0.5 = 1.060$. Wealth grows 6.0%.

1 Patient 202: Update counts—intervention has 101 patients, 35 events (34.7%); control has 100 patients, 41
2 events (41.0%). $\hat{\delta}_{201} = -0.063$. Patient 202 has an event. Bet toward control: $\lambda = 0.5 + 0.5 \times (-0.063) = 0.469$.
3 Assignment revealed: intervention. Wrong guess. Multiplier: $0.469/0.5 = 0.938$. Wealth **shrinks** 6.2%.
4 Cumulative wealth:

$$W_{202} = W_{199} \times 1.054 \times 1.060 \times 0.938 = W_{199} \times 1.048 \quad (16)$$

5 Despite one wrong guess, wealth grew 4.8% over these three patients. Under the alternative, correct guesses
6 outnumber incorrect ones on average and wealth grows. Under the null, right and wrong guesses balance out and
7 wealth fluctuates around 1.

8 3.5 Validity

9 The binary case is the simplest concrete instance of the unified validity argument. Here the hidden label is the
10 current treatment assignment, $A_i = T_i$, and the null probability is $\pi_i = p$. Under the null, outcome and treatment
11 are independent, so after observing Y_i and all prior data, the current treatment assignment still has probability
12 $P(T_i = 1) = p$. For any predictable wager λ_i ,

$$\mathbb{E}[\text{multiplier} \mid \mathcal{G}_i] = p \times \frac{\lambda_i}{p} + (1 - p) \times \frac{1 - \lambda_i}{1 - p} \quad (17)$$

$$= \lambda_i + (1 - \lambda_i) = 1 \quad (18)$$

13 Thus (W_i) is a nonnegative martingale under the null, and Ville’s inequality gives anytime-valid Type I error
14 control when rejecting at threshold $1/\alpha$.

15 3.6 Simulation studies

16 We evaluated operating characteristics of e-RTb by simulation. For each scenario, we calculated the sample size
17 required for a chi-square test to achieve the target power at $\alpha = 0.05$, then ran 5,000 simulated trials at that
18 sample size. We used burn-in = 50 patients and ramp = 100 patients. Control arm event rate was 40% in all
19 scenarios.

20 3.6.1 Results

21 Table 1 presents Type I error and power for trials designed to detect 5% or 10% absolute risk reductions (ARR)
22 with 80% or 90% power.

Table 1: Operating characteristics for adaptive e-RTb. Each row summarizes 5,000 fixed-seed simulated trials under the null and 5,000 under the matched alternative.

ARR	Target Power	N	Type I Error	e-RTb Power	Median Crossing
5%	80%	2,942	0.031	47.5%	1,450 (49%)
10%	80%	712	0.021	49.5%	401 (56%)
5%	90%	3,938	0.035	63.6%	1,837 (47%)
10%	90%	954	0.025	64.9%	479 (50%)

23 Type I error was controlled below the nominal $\alpha = 0.05$ level across the tested scenarios, consistent with the
24 martingale guarantee. Power was lower than the corresponding fixed-sample design power because the e-process
25 must cross an anytime-valid threshold. When the process rejected the null, it generally did so around the middle
26 of planned enrollment.

27 3.6.2 Trajectory examples

28 Figure 1 shows representative wealth trajectories from 30 simulated trials under the null hypothesis (both arms
29 40% event rate). Sample sizes correspond to trials designed for 80% power ($n = 712$, left) and 90% power (n

1 = 954, right) to detect a 10% ARR. Under the null, wealth fluctuates randomly around 1 and, in these panels,
2 gradually drifts downward as repeated bets are placed on noise.

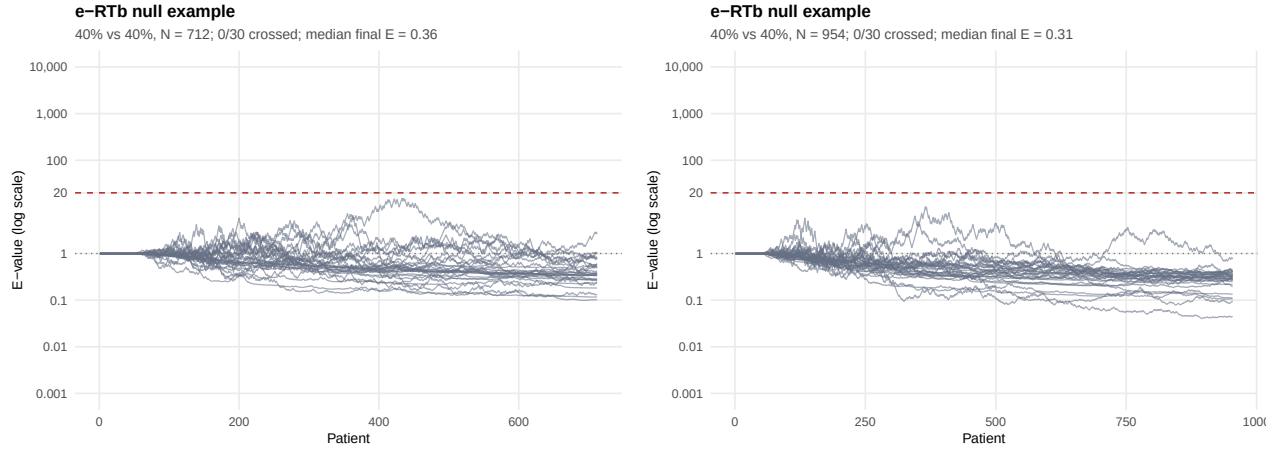


Figure 1: Wealth trajectories under the null hypothesis. Left: $n = 712$ (80% power design). Right: $n = 954$ (90% power design). Dashed red line: rejection threshold ($1/\alpha = 20$). Dotted gray line: neutral (wealth = 1). Under the null, no trajectory crosses the threshold.

3 Figure 2 shows representative trajectories under the alternative hypothesis (40% vs 30% event rates, true ARR
4 = 10%). With a real treatment effect, many wealth paths grow, cross the rejection threshold before enrollment
5 completes, and reach values far exceeding 20—providing strong evidence against the null.

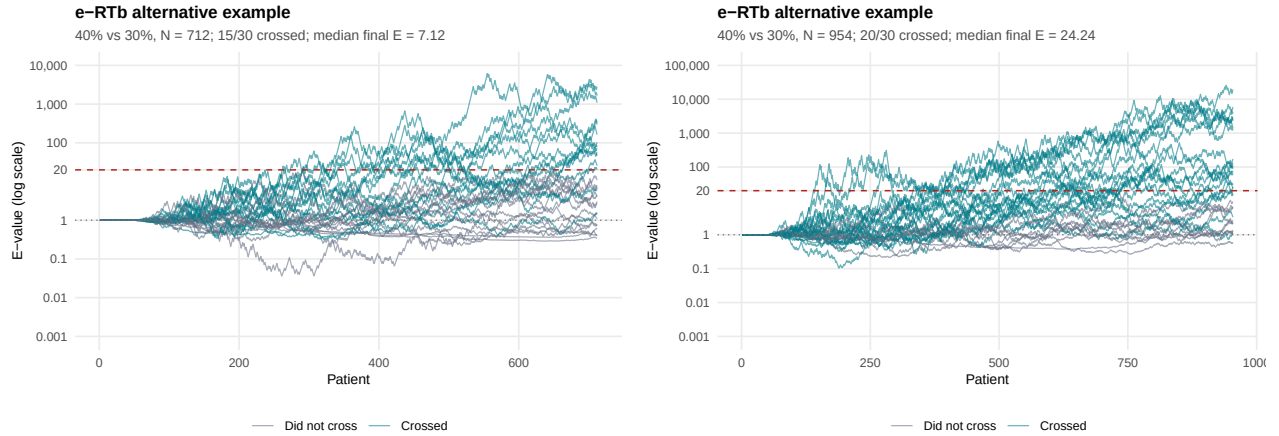


Figure 2: E-process trajectories under the alternative hypothesis (true ARR = 10%). Left: $n = 712$ (80% power design). Right: $n = 954$ (90% power design). Dashed red line: rejection threshold ($1/\alpha = 20$). Under the alternative, approximately half to two thirds of the representative trajectories cross the threshold, typically around the midpoint of enrollment.

6 4 Event-Only Monitoring (e-RTe)

7 4.1 Motivation

8 The e-RTb requires knowing both the treatment arm and the outcome for every enrolled patient. In practice,
9 this means a coordinator must ascertain whether each patient experienced the event or not—requiring follow-up,
10 data entry, and outcome adjudication for all patients, including the majority who do not experience the event.

1 In some settings, events of interest are reliably captured but non-events may not be systematically followed.
2 For example, in studies of mortality in acutely ill patients embedded in electronic medical records (EMR), a death
3 is an unmissable event; a survivor at day 28 requires active confirmation. Similarly, in oncology trials monitoring
4 disease progression, or in cardiovascular trials tracking myocardial infarction, the event is often captured with
5 high fidelity while confirming event-free status requires active follow-up. This motivates a simpler variant: **e-RTe**
6 **monitors only the stream of events, ignoring non-events entirely.**

7 4.2 Null Hypothesis

8 Consider a trial with 1:1 randomization. Under the null hypothesis of no treatment effect on the event rate, both
9 arms have the same event rate. If both arms have equal event rates and equal enrollment, then each event is
10 equally likely to come from either arm:

$$H_0 : P(\text{event from treatment} \mid \text{an event occurred}) = 0.5 \quad (19)$$

11 In plain language: if the treatment does not affect the event rate, then looking at the arm label on an event
12 is like flipping a fair coin. Treatment events and control events should arrive in roughly equal numbers.

13 This reduces trial monitoring to a sequential test of a Bernoulli coin: is the fraction of treatment events equal
14 to 0.5? We call this the “event coin.”

15 4.3 Algorithm

16 The algorithm maintains two counters: d_{trt} and d_{ctrl} , both initialized to zero. For each event $i = 1, 2, \dots$:

17 **Step 1: Estimate the coin bias from past data.** Using all events before the current one, compute the
18 plug-in estimator:

$$\hat{p}_{i-1} = \frac{d_{\text{trt}}}{d_{\text{trt}} + d_{\text{ctrl}}} \quad (20)$$

19 If no events have been observed yet, set $\hat{p} = 0.5$. In words: $\hat{p}$ is the running fraction of events that came from
20 the treatment arm. Under the null, this should hover around 0.5. Under the alternative (if treatment helps), this
21 should drift below 0.5.

22 **Step 2: Compute the wager.** The betting fraction is:

$$\lambda_i = 0.5 + c_i \cdot (\hat{p}_{i-1} - 0.5) \quad (21)$$

23 clamped to $[0.001, 0.999]$, where c_i is the same ramp function as before:

$$c_i = \min \left(1, \max \left(0, \frac{i - n_0}{n_r} \right) \right) \quad (22)$$

24 with burn-in $n_0 = 30$ events and ramp $n_r = 50$ events. During the burn-in, $c_i = 0$ and $\lambda_i = 0.5$ (a neutral
25 bet—no information yet). After the ramp completes, $c_i = 1$ and $\lambda_i = \hat{p}_{i-1}$: the wager fully reflects the observed
26 event proportion.

27 In plain language: we bet proportionally to what we have learned so far. If 40% of past events were from
28 treatment ($\hat{p} = 0.40$), we set $\lambda = 0.40$. This means we place more of our wager on the “control event” side, because
29 past data suggest control events are more common.

30 At full ramp ($c_i = 1$), this is the *full Kelly bet*—the wager that maximizes the expected logarithmic growth
31 rate of wealth. This contrasts with the half-Kelly strategy used in e-RTb. Full Kelly is viable here because events
32 are sparse: a trial with 2,000 patients and 15% event rate produces approximately 300 events. Over-betting
33 compounds across only a few hundred multiplications, causing gradual wealth erosion rather than the steeper
34 erosion seen with dense every-patient updates.

Step 3: Update wealth.

$$W_i = W_{i-1} \times \begin{cases} \lambda_i/0.5 & \text{if the event is from the treatment arm} \\ (1 - \lambda_i)/0.5 & \text{if the event is from the control arm} \end{cases} \quad (23)$$

starting from $W_0 = 1$.

In plain language: we place λ_i on “treatment event” and $1 - \lambda_i$ on “control event.” The null hypothesis pays 2 : 1 on the correct side (a fair payout for a 0.5-probability event). If we guessed correctly, wealth grows; if we guessed wrong, wealth shrinks.

Step 4: Update counters after betting. Increment d_{trt} or d_{ctrl} depending on which arm the event came from. This ordering—bet first, then update—ensures that the estimate $\hat{p}_{i-1}$ uses only past information, maintaining the martingale property.

4.4 Worked example

Consider a trial comparing a new treatment against standard of care in the ICU, where the event of interest is mortality. Suppose 80 deaths have been observed so far. Burn-in (30) and ramp (50) are complete, so $c_i = 1$. Current counts: $d_{\text{trt}} = 33$, $d_{\text{ctrl}} = 47$.

Event 81 arrives. Let us walk through the update.

Step 1: $\hat{p}_{80} = 33/(33 + 47) = 33/80 = 0.4125$. Interpretation: so far, 41.25% of events came from the treatment arm. This is below 50%, suggesting treatment may be protective.

Step 2: $\lambda_{81} = 0.5 + 1.0 \times (0.4125 - 0.5) = 0.4125$. We place 41.25% of our bet on “treatment event” and 58.75% on “control event.” We are leaning toward control because past evidence suggests treatment events are less frequent.

Step 3: The arm is revealed: it is a **control** event.

$$\text{Multiplier} = \frac{1 - 0.4125}{0.5} = \frac{0.5875}{0.5} = 1.175 \quad (24)$$

Wealth grows by 17.5%. Our bet was correct—we leaned toward control, and indeed it was a control event.

Step 4: Update $d_{\text{ctrl}} = 48$. Now $\hat{p}_{81} = 33/81 = 0.407$.

Had the event been from the treatment arm instead:

$$\text{Multiplier} = \frac{0.4125}{0.5} = 0.825 \quad (25)$$

Wealth would have shrunk by 17.5%. Our lean toward control would have been wrong.

Under the null, treatment and control events arrive with equal probability, so wins and losses balance on average. Under the alternative ($\hat{p} < 0.5$), control events are genuinely more frequent, and wealth grows systematically.

4.5 Validity

Theorem 2. *Under the null hypothesis $P(\text{event from treatment} \mid \text{event}) = 0.5$, the wealth process (W_i) is a nonnegative martingale.*

Proof. Condition on all past information and the current wager λ_i . Under the null, the probability that event i is from the treatment arm is exactly 0.5, independently of all past data. The expected wealth multiplier is:

$$\mathbb{E} \left[\frac{W_i}{W_{i-1}} \mid \lambda_i \right] = 0.5 \times \frac{\lambda_i}{0.5} + 0.5 \times \frac{1 - \lambda_i}{0.5} \quad (26)$$

$$= \lambda_i + (1 - \lambda_i) = 1 \quad (27)$$

This is the same identity as in the binary case. Since the expected multiplier is exactly 1, wealth cannot systematically grow under the null, regardless of how λ_i was chosen (as long as it depends only on past events). $\square$

1 By Ville’s inequality, $\Pr_{H_0}(\sup_{i \geq 1} W_i \geq 1/\alpha) \leq \alpha$. Rejecting when wealth crosses $1/\alpha$ controls Type I error
2 at any stopping time.

3 4.6 Adaptive bidirectionality

4 The adaptive e-RTe wager automatically detects both benefit and harm without pre-specifying a direction:

- 5 • If $\hat{p} < 0.5$ (fewer treatment events than expected): the method bets on control events being more common,
6 and wealth grows when treatment is indeed protective.
- 7 • If $\hat{p} > 0.5$ (more treatment events than expected): the method bets on treatment events being more common,
8 and wealth grows when treatment is harmful.

9 No investigator input about the expected direction is needed. The adaptive wager discovers the direction
10 from the data. The same principle also applies to adaptive e-RTb: if events become more common in treatment,
11 events point toward treatment; if events become less common in treatment, events point toward control. This
12 bidirectionality is a property of adaptive wagers. A design-fixed wager is instead directional unless a separate
13 two-sided or mirrored design rule is specified.

14 4.7 Signal Concentration

15 A key property of e-RTe is that it can outperform the full-sample e-RTb when the baseline event rate is low. The
16 intuition is that events *concentrate* the treatment signal.

17 Consider a trial where the control event rate is 25% and the treatment event rate is 20%, yielding a 5
18 percentage-point absolute risk reduction (ARR). In e-RTb, the signal per patient is diluted: most patients do not
19 experience the event, and only the 20–25% who do carry information about differential event rates. The observed
20 risk difference across all patients is 5 percentage points.

21 In e-RTe, only events are observed. The event-coin probability is:

$$p_{\text{alt}} = \frac{p_{\text{trt}}}{p_{\text{trt}} + p_{\text{ctrl}}} = \frac{0.20}{0.20 + 0.25} = 0.444 \quad (28)$$

22 This is an 11.2-point tilt from 0.5—more than double the 5-point ARR. The signal is concentrated because events
23 filter out the uninformative non-events.

24 This advantage diminishes as the baseline event rate increases. At higher event rates, e-RTb sees more
25 informative events per patient, and the event-coin tilt shrinks because both numerator and denominator grow:

Table 2: Signal concentration: event-coin tilt versus ARR for a 5pp risk reduction.				
Baseline Event Rate	Treatment Event Rate	Event Coin p_{alt}	Tilt from 0.5	Tilt / ARR
10%	5%	0.333	16.7 pp	3.33×
15%	10%	0.400	10.0 pp	2.00×
20%	15%	0.429	7.1 pp	1.43×
25%	20%	0.444	5.6 pp	1.11×
30%	25%	0.455	4.5 pp	0.91×
35%	30%	0.462	3.8 pp	0.77×
40%	35%	0.467	3.3 pp	0.67×

26 The crossover with e-RTb occurs at approximately 25% baseline event rate (Table 3). The analytical tilt
27 calculation places the boundary near 25%, while the finite-sample head-to-head simulation remains slightly e-
28 RTe-favorable at 25% and switches to e-RTb by 30%. Below that region, the concentrated event-coin signal more
29 than compensates for the smaller number of observations; above it, e-RTb’s access to all patients provides the
30 advantage.

4.8 Trajectory Examples

Figure 3 shows representative wealth trajectories for e-RTe. Under the null (left), the event coin is fair and wealth fluctuates near or below 1. Under the alternative (right), treatment events arrive less frequently than control events; in this 500-event illustration, 12 of 30 paths cross the threshold as the adaptive wager learns the event-coin imbalance.

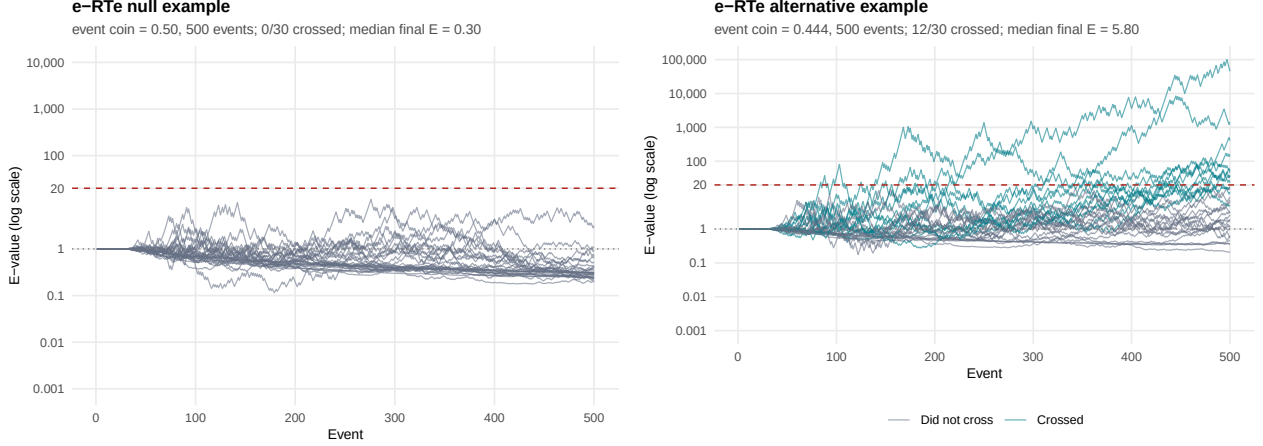


Figure 3: Trajectories of the e-RTe process (25% baseline event rate, 5pp ARR, 500 events). Left: under the null hypothesis (event coin = 0.50), wealth fluctuates randomly. Right: under the alternative hypothesis, some paths cross as the adaptive wager learns the event-coin imbalance. Dashed red line: rejection threshold ($1/\alpha = 20$).

4.9 Head-to-head comparison with e-RTb

To quantify the signal concentration crossover, we ran both e-RTe and e-RTb on the *same* simulated trials across a range of baseline event rates and absolute risk reductions (ARRs). For each scenario, we computed the frequentist sample size (two-proportion z -test, 80% power), enrolled that many patients, and analyzed the data with both methods: e-RTb processed all patients; e-RTe processed only the event stream. We used 2,000 simulations per scenario. Scenarios that would imply a treatment event rate below 5% were omitted.

For a 5pp ARR, the crossover occurs near 25% baseline event rate, consistent with the analytical prediction from Table 2. At 15–20% baseline, e-RTe outperforms e-RTb despite seeing fewer observations, because the event-coin tilt more than compensates for the smaller event stream. Above 30%, e-RTb’s access to all patients provides an increasingly large advantage. For larger ARR’s, the frequentist sample size shrinks and the event stream may become too short for e-RTe to complete its burn-in and ramp. In those scenarios, e-RTb usually dominates because the larger per-patient effect can be exploited immediately across all randomized patients.

Low-event constraints. At 10% baseline with a 5pp ARR, the frequentist sample size produces only about 66 expected events—fewer than the burn-in (30) plus ramp (50) = 80 events required for e-RTe to reach full betting strength. The e-RTe process does not complete its learning phase, so it cannot fully leverage the strong event-coin tilt. This highlights a practical constraint: e-RTe requires sufficient events to be effective.

Larger effects. With larger effects (7.5pp or 10pp ARR), the same pattern becomes more pronounced. The event-coin tilt is stronger, but the planned sample size becomes smaller, and e-RTe may have too few events for adaptive full-Kelly wagering to learn reliably before the trial ends.

Practical guidance. Use e-RTe when (i) the baseline event rate is low-to-moderate, (ii) the expected ARR is modest enough that the planned sample still yields a sufficiently long event stream, and (iii) ascertainment of non-events is impractical. When baseline event rates are high or effects are large enough to make the planned trial short, e-RTb is usually more powerful.

Table 3: Head-to-head comparison: e-RTe versus e-RTb across absolute risk reductions, using the same trial data and same enrolled-patient sample size. Scenarios requiring treatment event rates below 5% are omitted.

ARR	Baseline	Event Coin	N	Events	e-RTb Power	e-RTe Power	Δ	Winner
5.0 pp	10%	0.333	870	66	12.3%	8.1%	−4.2pp	e-RTb
5.0 pp	15%	0.400	1,372	172	24.8%	43.7%	+18.9pp	e-RTe
5.0 pp	20%	0.429	1,812	318	33.1%	44.7%	+11.6pp	e-RTe
5.0 pp	25%	0.444	2,188	493	40.0%	42.8%	+2.8pp	e-RTe
5.0 pp	30%	0.455	2,502	689	42.5%	37.4%	−5.1pp	e-RTb
5.0 pp	35%	0.462	2,754	896	46.0%	34.2%	−11.8pp	e-RTb
5.0 pp	40%	0.467	2,942	1,104	48.9%	32.1%	−16.8pp	e-RTb
7.5 pp	15%	0.333	556	63	19.7%	4.9%	−14.9pp	e-RTb
7.5 pp	20%	0.385	758	124	30.4%	35.5%	+5.1pp	e-RTe
7.5 pp	25%	0.412	932	199	40.5%	41.5%	+1.0pp	e-RTe
7.5 pp	30%	0.429	1,080	284	44.4%	37.0%	−7.4pp	e-RTb
7.5 pp	35%	0.440	1,198	375	49.0%	35.4%	−13.6pp	e-RTb
7.5 pp	40%	0.448	1,288	467	50.2%	33.4%	−16.9pp	e-RTb
10.0 pp	15%	0.250	282	29	10.8%	0.0%	−10.8pp	e-RTb
10.0 pp	20%	0.333	398	60	25.4%	2.8%	−22.6pp	e-RTb
10.0 pp	25%	0.375	500	100	33.1%	26.1%	−7.1pp	e-RTb
10.0 pp	30%	0.400	588	148	42.5%	33.3%	−9.2pp	e-RTb
10.0 pp	35%	0.417	658	198	46.2%	33.5%	−12.7pp	e-RTb
10.0 pp	40%	0.429	712	250	50.7%	34.2%	−16.4pp	e-RTb

4.10 Burn-in, ramp, and Kelly-intensity sensitivity

The previous comparison used the default adaptive settings: e-RTb used a 50-patient burn-in, 100-patient ramp, and half-intensity adaptive wager; e-RTe used a 30-event burn-in, 50-event ramp, and full-intensity adaptive wager. These defaults are reasonable but not sacred. Because e-RTe updates only at events, a fixed 30/50 event schedule may be too slow when the planned trial yields few events. Conversely, overly aggressive e-RTb betting can degrade wealth through many patient-level updates.

We therefore repeated the same- N comparison over a tuning grid. For each endpoint, we crossed fixed 10/20, fixed 30/50, default, proportional 5/10%, and proportional 10/20% burn-in/ramp schedules with 25%, 50%, 75%, and 100% adaptive Kelly intensity. Proportional schedules were defined on the natural update scale: planned patients for e-RTb and expected events for e-RTe. This is a sensitivity analysis rather than an optimized rule; choosing the best row after seeing trial data would not be a prespecified monitoring procedure. The resulting operating characteristics are summarized in Table 4.

The sensitivity analysis confirms that e-RTe performance is partly a tuning issue. Shorter or proportional ramps can recover substantial power when the default 30/50 event schedule consumes too much of the available event stream. The gain is most visible in short trials with strong event-coin tilt, where aggressive prespecified settings allow e-RTe to start betting before the trial is nearly over. This should be interpreted as design sensitivity, not as a data-adaptive tuning rule. As event streams become longer and baseline risk becomes higher, e-RTb often regains the advantage because it updates for every randomized patient and uses non-event information. e-RTb also benefits less from aggressive tuning because repeated patient-level over-betting can erode wealth.

5 Design-Calibrated Wagers for Binary and Event-Only Endpoints

The preceding simulations used the default wager policies: adaptive half-Kelly for e-RTb and adaptive full-Kelly for e-RTe. We explicitly separate the validity engine from the wager policy. The martingale argument requires only that the wager be predictable: it may be learned from prior trial data, or it may be fixed in advance from the design alternative. This distinction parallels Sokolova and Sokolov (2026a), where the e-process is calibrated to a clinically meaningful design effect.

A growth-rate-optimal (GROW) wager (Ramdas and Wang, 2025) is the value λ^* that maximizes the expected

Table 4: Sensitivity of adaptive e-RTb and e-RTe to burn-in/ramp schedule and Kelly intensity. Each row uses the same trial-data scenarios as Table 3; values summarize 1,000 fixed-seed simulations. Tuned columns show the best power over fixed 10/20-event, fixed 30/50-event, default, proportional 5/10%, and proportional 10/20% burn-in/ramp schedules crossed with 25%, 50%, 75%, and 100% Kelly intensity.

ARR	Baseline	Events	e-RTb Default	e-RTb Tuned	e-RTe Default	e-RTe Tuned	Winner
5.0 pp	10%	66	12.2%	19.6% (30/50 events, 100%K)	6.4%	48.3% (5/10%, 100%K)	e-RTe
5.0 pp	15%	172	25.2%	27.7% (5/10%, 75%K)	42.0%	48.3% (5/10%, 100%K)	e-RTe
5.0 pp	20%	318	32.9%	36.4% (10/20%, 75%K)	43.1%	43.1% (default, 100%K)	e-RTe
5.0 pp	25%	493	40.9%	41.2% (10/20%, 75%K)	42.6%	42.6% (default, 100%K)	e-RTe
5.0 pp	30%	689	43.8%	46.9% (10/20%, 75%K)	37.2%	38.0% (5/10%, 100%K)	e-RTb
5.0 pp	35%	896	47.9%	51.7% (10/20%, 75%K)	36.1%	38.8% (5/10%, 100%K)	e-RTb
5.0 pp	40%	1,104	46.4%	51.0% (5/10%, 50%K)	29.3%	30.1% (5/10%, 100%K)	e-RTb
7.5 pp	15%	63	20.6%	27.0% (default, 75%K)	6.6%	46.6% (10/20%, 100%K)	e-RTe
7.5 pp	20%	124	31.6%	34.0% (10/20%, 75%K)	32.5%	46.1% (10/20 events, 100%K)	e-RTe
7.5 pp	25%	199	37.9%	40.0% (10/20%, 75%K)	38.5%	40.2% (10/20%, 100%K)	~Tied
7.5 pp	30%	284	42.3%	44.2% (10/20%, 75%K)	38.6%	38.9% (10/20%, 100%K)	e-RTb
7.5 pp	35%	375	46.6%	46.6% (default, 50%K)	35.3%	36.1% (5/10%, 100%K)	e-RTb
7.5 pp	40%	467	51.8%	53.1% (10/20%, 75%K)	35.9%	36.6% (10/20%, 100%K)	e-RTb
10.0 pp	15%	29	13.5%	25.5% (default, 100%K)	0.0%	46.8% (5/10%, 75%K)	e-RTe
10.0 pp	20%	60	22.2%	31.0% (default, 100%K)	2.7%	41.9% (10/20%, 100%K)	e-RTe
10.0 pp	25%	100	37.1%	41.4% (default, 75%K)	25.2%	43.4% (10/20 events, 100%K)	e-RTe
10.0 pp	30%	148	42.4%	42.6% (10/20%, 75%K)	37.5%	43.9% (10/20 events, 100%K)	e-RTe
10.0 pp	35%	198	45.9%	48.2% (10/20%, 75%K)	34.6%	36.7% (10/20 events, 100%K)	e-RTb
10.0 pp	40%	250	51.3%	52.2% (10/20%, 75%K)	33.6%	34.0% (10/20%, 100%K)	e-RTb

1 log-growth of the e-process under a specified design alternative:

$$\lambda^* = \arg \max_{\lambda} \mathbb{E}_{\text{design}} \{\log M(\lambda)\}, \quad (29)$$

2 where $M(\lambda)$ is the one-step e-process multiplier. In the binary paired-comparison construction of Sokolova and
3 Sokolov (2026a), this wager is computed from the design alternative before monitoring and then held fixed during
4 the e-value run. This is analogous to a frequentist design effect used for sample-size planning: it improves efficiency
5 when the design alternative is close to the truth, but can lose power when the effect is misspecified. The design-
6 fixed e-RT policies below adopt the same planning philosophy, while retaining the e-RT assignment-prediction
7 construction rather than the paired-comparison construction.

8 For e-RTb, the design-fixed wager uses the posterior assignment probabilities implied by prespecified event
9 rates. With 1:1 randomization and design rates p_T and p_C ,

$$\lambda_{\text{event}} = \Pr(T = 1 \mid Y = 1) = \frac{p_T}{p_T + p_C}, \quad \lambda_{\text{non-event}} = \Pr(T = 1 \mid Y = 0) = \frac{1 - p_T}{(1 - p_T) + (1 - p_C)}. \quad (30)$$

10 For e-RTe, the design-fixed wager is the corresponding event-coin probability, $\Pr(T = 1 \mid \text{event}) = p_T / (p_T + p_C)$.
11 Thus e-RTb fixes both an event and non-event wager, whereas e-RTe fixes only the event wager.

12 We ran 5,000 simulations per scenario using a fixed seed. Sample sizes were anchored to the usual fixed-sample
13 two-proportion calculation in R (`power.prop.test`, 80% power, $\alpha = 0.05$). Both e-RTb and e-RTe were evaluated
14 at the same enrolled-patient N ; no event-only inflation was used in this comparison. Under the null, we reused
15 the design N for 5pp and 10pp ARR alternatives. Under the alternative, we compared adaptive wagering against
16 fixed wagers that underestimated, matched, or overestimated the true ARR. An oracle full-Kelly row is included
17 only as a simulation benchmark. Type I error and power are reported in Tables 5 and 6.

18 The adaptive null rows in Table 5 are now directly comparable in precision to the 5,000-replicate adaptive
19 e-RTb baseline in Table 1. At the 10pp design sample size, the adaptive e-RTb null estimate is 2.0% here and
20 2.1% in the baseline run, supporting the interpretation that the lower Type I error in the shorter 10pp-design
21 trial reflects the operating scenario rather than a separate setup difference.

22 The key result is that design-fixed full-Kelly wagers can substantially increase power when the design effect is
23 close to the truth, while retaining Type I error control in these simulations. However, misspecification matters.
24 Underestimating the effect is usually conservative; overestimating the effect can produce earlier crossings among
25 successful trials but lower overall power, especially for e-RTe at a true 5pp ARR. Because e-RTe observes only
26 events, the same enrolled-patient N produces fewer betting opportunities than e-RTb. Overall, design-fixed wagers

Table 5: Type I error for adaptive and design-fixed wager policies at the same enrolled-patient sample sizes. Each row summarizes 5,000 simulated null trials with $p_C = 0.40$ and $\alpha = 0.05$. Sample sizes were obtained from the usual fixed-sample two-proportion calculation with 80% power; no event-only inflation was used for e-RTe in this comparison.

Endpoint	Scenario	Policy	Wager ARR	N	Events	Type I
e-RTb	Null 5.0pp design	Adaptive	–	2,942	–	3.5%
e-RTb	Null 5.0pp design	Fixed 5pp	5.0pp	2,942	–	3.7%
e-RTb	Null 5.0pp design	Fixed 10pp	10.0pp	2,942	–	4.5%
e-RTb	Null 10.0pp design	Adaptive	–	712	–	2.0%
e-RTb	Null 10.0pp design	Fixed 5pp	5.0pp	712	–	0.3%
e-RTb	Null 10.0pp design	Fixed 10pp	10.0pp	712	–	3.3%
e-RTe	Null 5.0pp design	Adaptive	–	2,942	1,177	3.2%
e-RTe	Null 5.0pp design	Fixed 5pp	5.0pp	2,942	1,177	2.7%
e-RTe	Null 5.0pp design	Fixed 10pp	10.0pp	2,942	1,177	4.8%
e-RTe	Null 10.0pp design	Adaptive	–	712	285	1.7%
e-RTe	Null 10.0pp design	Fixed 5pp	5.0pp	712	285	0.1%
e-RTe	Null 10.0pp design	Fixed 10pp	10.0pp	712	285	2.0%

Table 6: Power for adaptive and design-fixed wager policies at the same enrolled-patient sample sizes. Each row summarizes 5,000 simulated trials with $p_C = 0.40$ and $\alpha = 0.05$. Fixed policies use full-Kelly wagers calibrated to the listed wager ARR; adaptive e-RTb uses half-Kelly and adaptive e-RTe uses full-Kelly.

Endpoint	Scenario	Policy	Wager ARR	N	Events	Power	Median crossing
e-RTb	True 5.0pp	Adaptive	–	2,942	–	49.3%	1,452
e-RTb	True 5.0pp	Fixed under	2.5pp	2,942	–	53.7%	2,154
e-RTb	True 5.0pp	Fixed matched	5.0pp	2,942	–	75.0%	1,438
e-RTb	True 5.0pp	Fixed over	10.0pp	2,942	–	55.7%	824
e-RTb	True 5.0pp	Oracle	5.0pp	2,942	–	73.9%	1,453
e-RTb	True 10.0pp	Adaptive	–	712	–	50.5%	396
e-RTb	True 10.0pp	Fixed under	5.0pp	712	–	41.9%	565
e-RTb	True 10.0pp	Fixed matched	10.0pp	712	–	71.3%	404
e-RTb	True 10.0pp	Fixed over	15.0pp	712	–	67.1%	327
e-RTb	True 10.0pp	Oracle	10.0pp	712	–	71.4%	409
e-RTe	True 5.0pp	Adaptive	–	2,942	1,104	31.5%	556
e-RTe	True 5.0pp	Fixed under	2.5pp	2,942	1,104	14.2%	936
e-RTe	True 5.0pp	Fixed matched	5.0pp	2,942	1,104	51.2%	695
e-RTe	True 5.0pp	Fixed over	10.0pp	2,942	1,104	46.9%	404
e-RTe	True 5.0pp	Oracle	5.0pp	2,942	1,104	52.2%	686
e-RTe	True 10.0pp	Adaptive	–	712	250	33.8%	158
e-RTe	True 10.0pp	Fixed under	5.0pp	712	250	5.7%	227
e-RTe	True 10.0pp	Fixed matched	10.0pp	712	250	43.2%	182
e-RTe	True 10.0pp	Fixed over	15.0pp	712	250	50.6%	145
e-RTe	True 10.0pp	Oracle	10.0pp	712	250	42.4%	181

1 appear useful as an optional mode rather than as a replacement for adaptive effect-size-agnostic monitoring.
 2 Figures 4 and 5 show the corresponding operating-characteristic plots, and Figure 6 shows representative wealth
 3 paths under the 5pp ARR scenario.

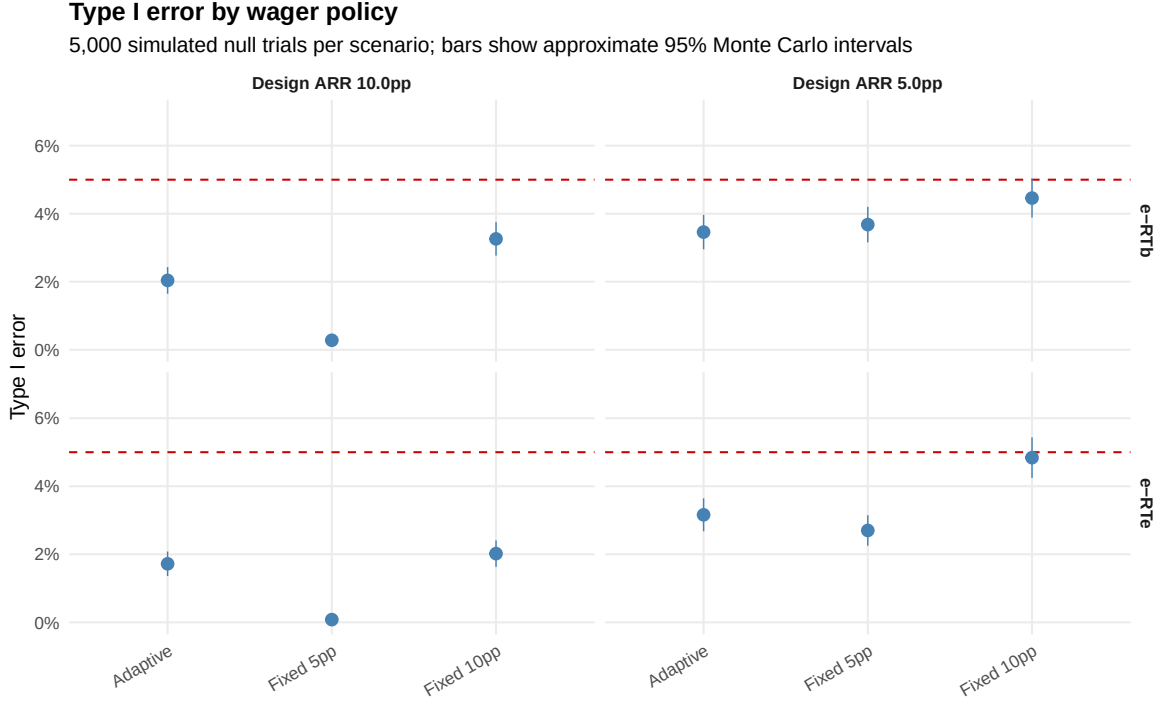


Figure 4: Type I error for adaptive and design-fixed wager policies. Each point summarizes 5,000 null simulations. The dashed red line is the nominal $\alpha = 0.05$ threshold.

4 6 Effect Estimates at Crossing and Type M Error

5 Because e-RT methods may stop early, the apparent treatment effect at the first threshold crossing is expected
 6 to be inflated. Following the design-analysis terminology of Gelman and Carlin (2014), this is a form of Type M
 7 error: the magnitude of the observed effect among trials that cross may exceed the true effect because crossings
 8 are enriched for favorable random fluctuations. The companion diagnostic is Type S error, the probability that
 9 a selected or crossing estimate points in the wrong direction. This does not invalidate the e-value—the e-process
 10 itself remains anytime-valid—but it matters for clinical interpretation. A trial stopped at an e-value threshold
 11 should not interpret the naive effect estimate at crossing as an unbiased estimate of the final treatment effect.

12 We therefore summarized the apparent effect at first crossing among simulated trials that crossed. For both
 13 e-RTb and e-RTe, the displayed effect scale is the apparent absolute risk reduction using all randomized patients
 14 observed by the time of first crossing. For e-RTe, this is a diagnostic snapshot rather than information used by the
 15 event-only e-process: the e-RTe wealth process itself still sees only event arm labels, but when denominators are
 16 available at the crossing time, the clinical effect size can be estimated on the same scale as e-RTb. If denominators
 17 are not available operationally, only the native event-coin diagnostic can be reported. Table 7 reports the binary
 18 and event-only crossing diagnostics; analogous scale-specific Type M diagnostics are reported below for e-RTc and
 19 e-RTs.

20 The inflation is clinically meaningful. In the 5pp ARR scenario, adaptive e-RTb crossings had a median
 21 apparent ARR of 7.92 percentage points, with a median Type M ratio of 1.58. Adaptive e-RTe crossings had a
 22 similar full-data snapshot ARR of 8.03 percentage points, with a median Type M ratio of 1.61. Fixed matched
 23 wagers reduced this inflation for both methods, with median ratios of 1.30 for e-RTb and 1.35 for e-RTe. These
 24 results reinforce that e-RT crossings should be interpreted as valid evidence for a treatment difference, not as

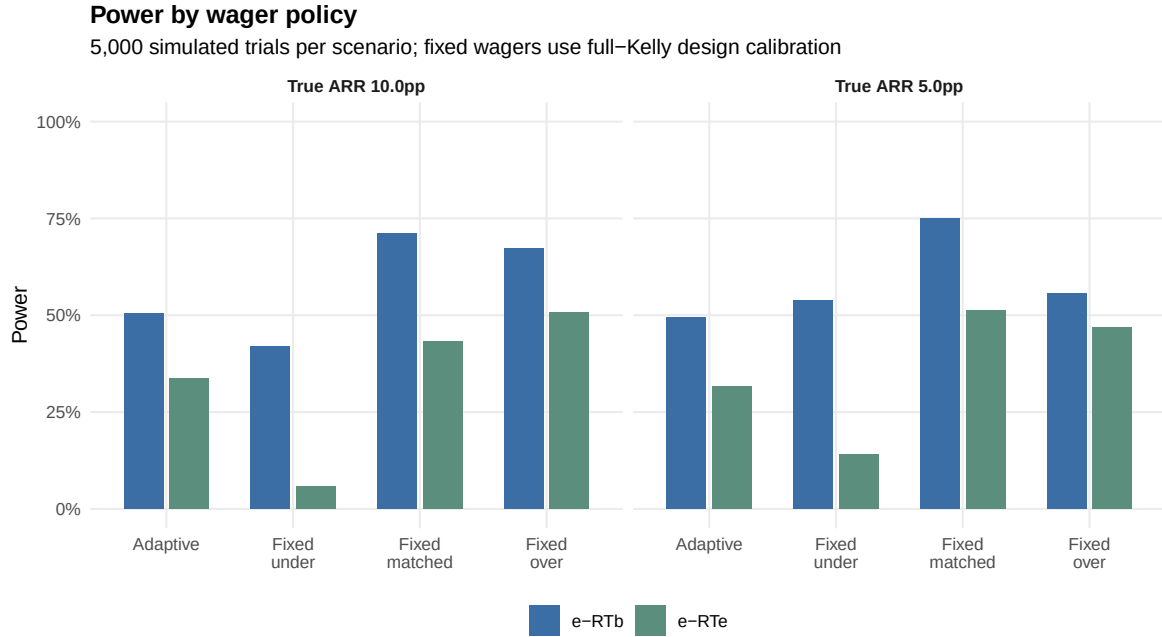


Figure 5: Power for adaptive and design-fixed wager policies. Fixed matched wagers are calibrated to the true ARR; underestimated and overestimated wagers deliberately misspecify the design effect.

Table 7: Type M error at first e-process crossing for e-RTb and e-RTe. Crossing and final effects are absolute risk reductions in percentage points. For e-RTe, the e-process itself remains event-only; the absolute risk reduction is a full-data diagnostic computed from all randomized patients observed by the crossing time. Type M is computed among trials that crossed.

True	Endpoint	Policy	Crossing	Final	Median M	Q75	Q90
5.0pp	e-RTb	Adaptive	7.92	5.00	1.58	2.12	2.95
5.0pp	e-RTb	Fixed matched	6.52	5.01	1.30	1.69	2.15
5.0pp	e-RTe	Adaptive	8.03	4.99	1.61	2.03	2.66
5.0pp	e-RTe	Fixed matched	6.73	4.99	1.35	1.64	2.00
10.0pp	e-RTb	Adaptive	14.68	10.06	1.47	1.79	2.20
10.0pp	e-RTb	Fixed matched	12.74	10.03	1.27	1.57	1.89
10.0pp	e-RTe	Adaptive	14.63	10.09	1.46	1.76	2.11
10.0pp	e-RTe	Fixed matched	13.15	10.02	1.32	1.58	1.83

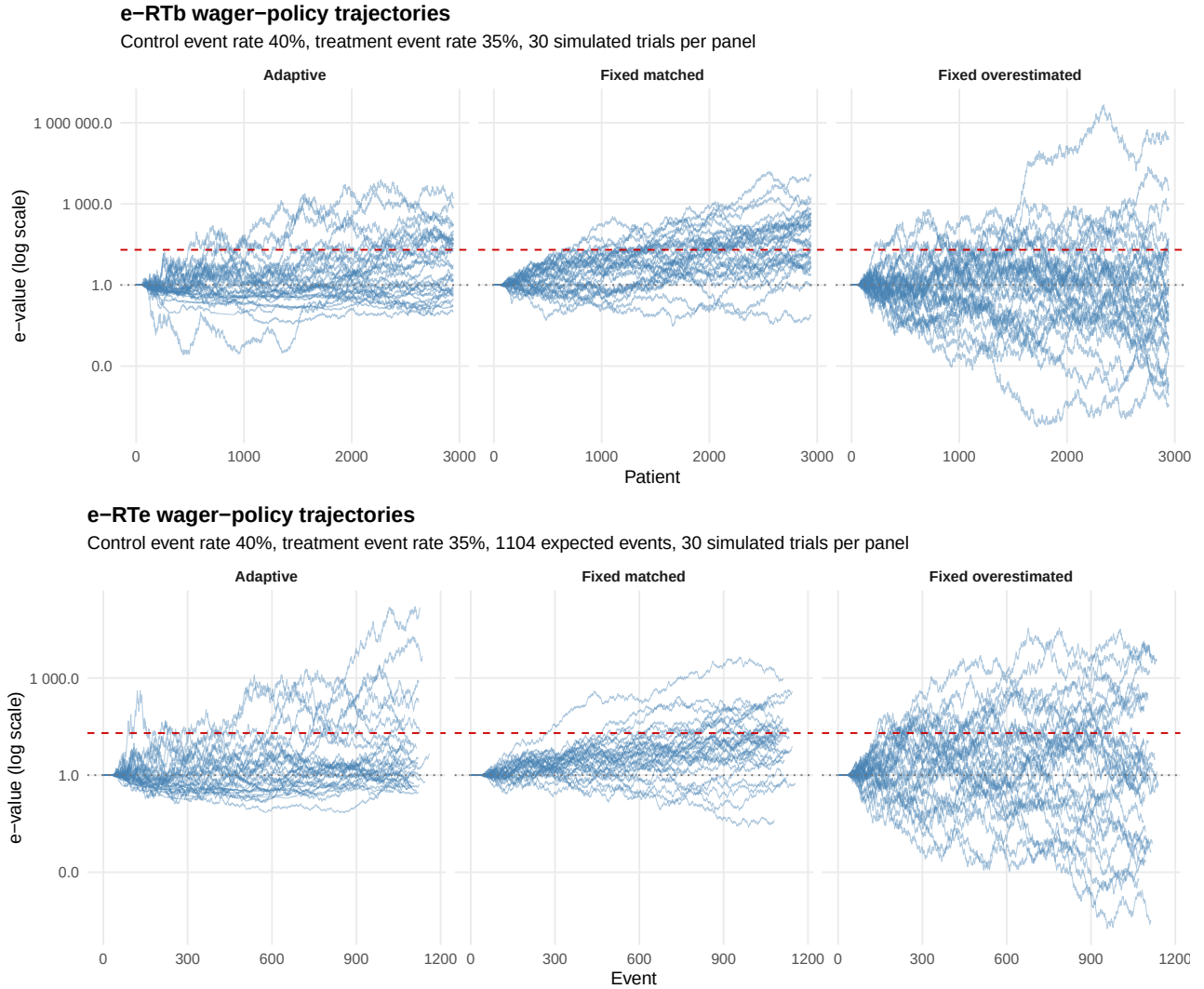


Figure 6: Example wealth trajectories under a true 5pp ARR with control event rate 40%. Top: e-RTb using all enrolled patients. Bottom: e-RTe using only events. Each panel shows 30 simulated trials.

1 unbiased estimates of its magnitude.

2 7 Continuous Outcomes

3 The e-RTb treats each patient as a single Bernoulli trial: the outcome (event vs. no event) is observed, and we
4 bet on which arm that patient came from. For continuous endpoints, the logic is the same but the signal is richer.
5 Each patient now contributes a continuous measurement (for example, ventilator-free days, change in biomarker,
6 or a physiologic score), and the betting strategy uses how extreme that value is relative to past data. Therefore,
7 the wager definition is slightly more granular.

8 We call the continuous endpoint extension e-RTc. The validity still comes only from randomization: under
9 the null hypothesis, the distribution of the continuous outcome is the same in both arms, so the outcome does
10 not help to predict treatment assignment.

11 7.1 Setup

12 At each enrollment $i = 1, 2, \dots$, the trial generates two pieces of information. First, the randomization mechanism
13 assigns the patient to one of the two arms, denoted by $T_i \in \{0, 1\}$ (with 0 for control and 1 for intervention),
14 using a known allocation probability p (typically $p = 0.5$ for 1:1 randomization, although equal allocation is not
15 required). Second, once follow-up is complete, we observe a continuous outcome $Y_i \in \mathbb{R}$ such as ventilator-free
16 days, a biomarker concentration, or a physiologic measurement. So far, same as before but using a continuous
17 endpoint.

18 Under the null hypothesis of no treatment effect, the distribution of Y_i is identical in both arms, and hence
19 the outcome carries no information about the treatment assignment; formally, $Y_i \perp T_i$ under H_0 . This single
20 independence relationship is the foundation of the continuous randomization e-process. The idea mirrors the
21 binary approach: each patient creates a small betting game in which we observe Y_i , form a data-driven guess
22 about which arm the patient is more likely to belong to, and then update our wealth once the actual assignment
23 T_i is revealed. If the null is true, these guesses cannot systematically win because outcomes are uninformative
24 about treatment. If the alternative is true, outcomes begin to separate between arms, the bets gain predictive
25 power, and wealth grows accordingly.

26 Viewed this way, extending the randomization e-process from binary to continuous outcomes does not require
27 additional assumptions for validity; it replaces the event/no-event signal with a continuous measure of extremeness
28 relative to past observations, while preserving the anytime-valid martingale structure. The difference lies in how
29 the wager is defined.

30 7.2 Betting strategy for continuous outcomes

31 We can use a data-driven rule with three ingredients:

- 32 1. A *center* and *scale* for the outcomes observed so far. Outcomes can be all over the place. We need to
33 standardize.
- 34 2. A standardized residual for the new outcome. As the reader will see we bet proportionally to how far the
35 result is away from the center reference we chose. This also needs to be standardized.
- 36 3. A smooth map from that residual into a betting fraction $\lambda_i \in (0, 1)$. If the measurement is an outlier, a
37 huge wager would be placed because difference between measurement and the center of the scale would be
38 huge, we need to muffle it to something between $(0, 1)$.

39 In simpler terms: Each new outcome is first compared to the past outcomes to understand how “unusual” it
40 is. To do this, we anchor the outcome to a robust center and a robust scale: the median gives the center and
41 the MAD (median absolute deviation) gives the spread. We then compute a standardized value: how far above
42 or below the median this new observation is, measured in MAD units. That standardized number is what guides
43 how aggressively we bet. A large positive value means “this looks unusually high compared with past outcomes,”

1 a large negative value means the opposite, and values near zero mean “this one looks typical.” The point of the
2 MAD is simple: it behaves well even when early data are messy or skewed. It stops a single extreme value from
3 blowing up the bet and keeps the e-process stable while the trial is still young. This likely comes at the cost of
4 reduced power.

5 At step i , using all previous outcomes $Y_1, \dots, Y_{i-1}$, we compute:

$$m_{i-1} = \text{median}(Y_1, \dots, Y_{i-1}), \quad (31)$$

$$s_{i-1} = \text{MAD}(Y_1, \dots, Y_{i-1}), \quad (32)$$

6 where MAD is the median absolute deviation,

$$\text{MAD}(Y_1, \dots, Y_{i-1}) = \text{median}_{1 \leq \ell < i} |Y_\ell - m_{i-1}|. \quad (33)$$

7 The median and MAD are robust to outliers and skewness. If s_{i-1} is zero or not finite, we set $s_{i-1} = 1$ to avoid
8 degeneracy.

9 For the new patient, we form a standardized residual:

$$r_i = \frac{Y_i - m_{i-1}}{s_{i-1}}. \quad (34)$$

10 This means that the patient residual r_i is the observed value Y_i minus the median observed so far, divided
11 by the MAD s_{i-1} . Note how this uses only outcomes observed before patient i for centering and scaling, keeping
12 the wager predictable.

We then squash this standardized value into the interval $(-1, 1)$ using

$$g_i = \frac{r_i}{1 + |r_i|}.$$

13 This is a simple monotone transformation: for moderate values $g_i \approx r_i$, while very large positive or negative
14 residuals are shrunk toward $+1$ or -1 . The only purpose of this step is to prevent a single extreme observation
15 from forcing an almost all-in bet.

16 Next, we ramp up the betting strength over time. Let

$$c_i = \min \left\{ 1, \max \left(0, \frac{i - \text{burn-in}}{\text{ramp}} \right) \right\}, \quad (35)$$

17 where **burn-in** is the number of initial patients during which we essentially do not bet, and **ramp** controls how
18 quickly we move from very cautious betting to our maximum aggressiveness. Those concepts are exactly like the
19 binary approach. Finally, we cap the maximum betting strength at $c_{\max} \in (0, 0.5]$ to avoid pathological bets.

20 Additionally, we need a direction estimate: which arm has better outcomes? The default adaptive rule keeps
21 this deliberately coarse. Using all previous data, we compute

$$q_{i-1} = \text{sign}(\bar{Y}_{\text{trt}, i-1} - \bar{Y}_{\text{ctrl}, i-1}), \quad (36)$$

22 with $q_{i-1} = 0$ if either arm has no prior observations. This makes adaptive e-RTc effect-size agnostic: the running
23 data choose the direction of the bet, but not a prespecified effect magnitude. The magnitude of the wager still
24 depends on the current observation-level score g_i and the ramped betting strength.

25 The betting fraction λ_i is then

$$\lambda_i = 0.5 + c_i \cdot c_{\max} \cdot g_i \cdot q_{i-1}. \quad (37)$$

26 By construction, $\lambda_i \in (0, 1)$ and is predictable: it depends only on past outcomes and the new Y_i , not on T_i . In
27 words: if the current observation Y_i is extreme in the direction that past data associate with the treatment arm,
28 λ_i deviates substantially from 0.5, placing a confident bet on treatment. If Y_i is typical, or if past data have not
29 yet identified a direction ($q_{i-1} = 0$), λ_i stays near 0.5 and we barely bet.

30 Intuitively:

- If Y_i is close to the historical median, $g_i \approx 0$ and $\lambda_i \approx 0.5$: we essentially do not bet, regardless of how strong the estimated treatment effect is.
- If Y_i is extreme (large $|g_i|$) and past data suggest a treatment-favorable direction, then λ_i deviates substantially from 0.5: we place a confident bet on one arm. The product $g_i \cdot q_{i-1}$ determines the direction, while $|g_i|$ and $c_i c_{\max}$ determine magnitude.
- If Y_i is extreme but $q_{i-1} = 0$ (no estimated direction yet), λ_i stays near 0.5—an unusual observation is not informative if we do not yet know which direction to bet.
- Early in the trial, c_i is small, so even unusual observations lead to mild bets. As data accumulates, c_i approaches 1 and the bets become more confident.

7.3 Parametric design wager for continuous outcomes

The adaptive wager above is intentionally agnostic to the effect size. For trials with a credible design alternative, e-RTc can also use a parametric design wager. Suppose the protocol specifies a normal-shift working model,

$$Y \mid T = 0 \sim N(\mu_C^*, \sigma^{*2}), \quad Y \mid T = 1 \sim N(\mu_T^*, \sigma^{*2}).$$

After observing the new outcome Y_i , but before revealing or using T_i in the betting update, the design wager is

$$\lambda_i^* = \Pr_{\text{design}}(T_i = 1 \mid Y_i) = \frac{p f_1^*(Y_i)}{p f_1^*(Y_i) + (1-p) f_0^*(Y_i)},$$

where f_1^* and f_0^* are the design densities for treatment and control. As before, we may ramp from the neutral wager p toward λ_i^* during early enrollment.

This design wager is parametric by construction. The normal-shift model is not needed for validity; it only determines how aggressively the e-process bets. Under the null, treatment assignment remains randomized and independent of Y_i , so any predictable $\lambda_i(Y_i)$ gives an expected wealth multiplier of 1. Misspecifying μ_C^* , μ_T^* , or σ^* can reduce power or increase Type M error at crossing, but it does not break the martingale guarantee.

7.4 Wealth update

Treatment is still randomized with probability p of intervention. After we choose λ_i , the wealth updates exactly as in the binary approach:

$$W_i = W_{i-1} \times \begin{cases} \lambda_i/p & \text{if } T_i = 1 \text{ (intervention)} \\ (1 - \lambda_i)/(1 - p) & \text{if } T_i = 0 \text{ (control)} \end{cases} \quad (38)$$

with $W_0 = 1$.

The only difference from the binary case is how we choose λ_i , as we saw. For binary outcomes, λ_i depends on the event indicator and past event rates by arm. For continuous outcomes, λ_i depends on how extreme Y_i is relative to past outcomes.

7.5 Worked intuition

Imagine a trial where the outcome is ventilator-free days, and higher is better. Suppose that after 100 patients, the median and MAD of Y are roughly stable, and the treatment arm has had better outcomes so far, so $q_{100} = 1$. Patient 101 has an unusually high number of ventilator-free days compared with this distribution. The standardized residual r_{101} is positive and large, so $g_{101} \approx 0.8$ and, after burn-in, $c_{101} \approx 1$. With $c_{\max} = 0.6$, we get $\lambda_{101} \approx 0.5 + 0.6 \times 0.8 \times 1 \approx 0.98$: we strongly bet that this patient was in the intervention arm. If they indeed were, wealth increases by roughly a factor of $0.98/0.5 \approx 2$ for this one patient. If not, wealth shrinks by $(1 - 0.98)/0.5 \approx 0.04$.

Under the null, high values like this are just as likely in control as in intervention: we win and lose in balance, and wealth does not grow on average. Under a true benefit, such favorable outliers cluster in the intervention arm, and the bets pay off more often than not.

7.6 Validity

The key point is that validity does not depend on the choice of median, MAD, or the specific transformation g_i . It depends only on the fact that:

1. λ_i is chosen *before* observing T_i and depends only on past data and Y_i ;
2. under the null, T_i is independent of Y_i with $\mathbb{P}(T_i = 1) = p$.

Theorem 3. *Under the null hypothesis of no treatment effect, the e-process wealth process (W_i) is a test martingale: for all i ,*

$$\mathbb{E}[W_i \mid \mathcal{F}_{i-1}] \leq W_{i-1},$$

where $\mathcal{F}_{i-1}$ is the sigma-field generated by all observations up to step $i - 1$.

Proof. Condition on $\mathcal{F}_{i-1}$ and Y_i . The bet λ_i is now fixed. Under the null, T_i is independent of Y_i and

$$\mathbb{P}(T_i = 1 \mid \mathcal{F}_{i-1}, Y_i) = p, \quad \mathbb{P}(T_i = 0 \mid \mathcal{F}_{i-1}, Y_i) = 1 - p.$$

The conditional expectation of the wealth multiplier is:

$$\mathbb{E}\left[\frac{W_i}{W_{i-1}} \mid \mathcal{F}_{i-1}, Y_i\right] = p \cdot \frac{\lambda_i}{p} + (1 - p) \cdot \frac{1 - \lambda_i}{1 - p} \quad (39)$$

$$= \lambda_i + (1 - \lambda_i) \quad (40)$$

$$= 1. \quad (41)$$

Thus $\mathbb{E}[W_i \mid \mathcal{F}_{i-1}, Y_i] = W_{i-1}$, and taking expectations over Y_i yields $\mathbb{E}[W_i \mid \mathcal{F}_{i-1}] = W_{i-1}$. This shows that (W_i) is a martingale with unit expectation under the null. $\square$

As in the binary case, Ville's inequality implies that for any stopping time τ ,

$$\mathbb{P}(W_\tau \geq 1/\alpha) \leq \alpha,$$

so rejecting the null when $W_\tau \geq 1/\alpha$ controls Type I error at level α , regardless of the stopping rule.

7.7 Simulation overview

We evaluated e-RTc using the same design philosophy as the binary and event-only simulations. For a given standardized effect size (Cohen's d), we first computed the fixed-sample size required for a two-sample t -test with 80% power at $\alpha = 0.05$. We then simulated 1,000 trials per scenario with 1:1 randomization and normally distributed outcomes with common standard deviation 1. The adaptive e-RTc used the V7 sign-direction wager with burn-in = 20, ramp = 50, and $c_{\max} = 0.6$. The design e-RTc used the normal-shift design wager above, with matched, underestimated, and overestimated design effects. Table 8 reports the null simulations, and Table 9 reports power and Type M behavior under alternatives.

Type I error remained controlled in these simulations. The adaptive sign-direction wager is conservative for small effects: at true $d = 0.20$, it crossed in only 9.8% of trials and did so very early when it crossed, producing a high median Type M ratio of 2.98. The parametric design wager improved power substantially when the design effect was close to the truth: at $d = 0.20$, matched design wagering increased power to 73.4% and reduced median Type M to 1.28. The tradeoff is visible under misspecification. Overestimating the design effect generally caused earlier crossings and more Type M inflation, while underestimating the effect delayed crossings and sometimes reduced power. These results support design-calibrated e-RTc as an optional efficiency mode, not as a replacement for the adaptive effect-size-agnostic monitor.

Table 8: Type I error for e-RTc adaptive and parametric design wagers. Each row summarizes 1,000 simulated null trials. Sample sizes use the usual fixed-sample two-sample t -test with 80% power and $\alpha = 0.05$.

Design d	Policy	N	Type I	Median final e-value
0.20	Adaptive	788	3.8%	0.00
0.20	Design	788	3.0%	0.03
0.40	Adaptive	200	4.3%	0.00
0.40	Design	200	2.9%	0.05
0.60	Adaptive	90	4.0%	0.00
0.60	Design	90	1.2%	0.21

Table 9: Power and Type M error for e-RTc adaptive and parametric design wagers. The effect scale is Cohen’s d ; Type M is computed among trials that crossed.

True d	Policy	Wager d	N	Power	Median crossing	Crossing d	Median M
0.20	Adaptive	–	788	9.8%	64	0.60	2.98
0.20	Design under	0.10	788	52.1%	587	0.27	1.33
0.20	Design matched	0.20	788	73.4%	401	0.26	1.28
0.20	Design over	0.40	788	59.2%	252	0.32	1.58
0.40	Adaptive	–	200	31.6%	67	0.67	1.66
0.40	Design under	0.20	200	34.4%	168	0.56	1.39
0.40	Design matched	0.40	200	66.6%	130	0.51	1.29
0.40	Design over	0.80	200	61.1%	93	0.55	1.37
0.60	Adaptive	–	90	53.8%	62	0.77	1.28
0.60	Design under	0.30	90	5.7%	85	0.96	1.60
0.60	Design matched	0.60	90	44.7%	77	0.78	1.29
0.60	Design over	1.20	90	55.5%	67	0.75	1.25

A visual representation of the matched normal-shift design wager is shown in Figure 7.

Figure 7 illustrates 30 simulated trajectories of the process for continuous endpoints under a design targeting a standardized effect size of $d = 0.40$ with 80% power. Under the null hypothesis (left panel), the prespecified design wager tends to spend wealth against noise, and no trajectory crosses the rejection threshold of $1/\alpha = 20$ in this representative panel. Under the corresponding alternative (right panel), outcomes separate between arms, the design wager is systematically favorable, and many trajectories cross before the planned sample size, demonstrating the potential for early stopping.

8 Time-to-Event Outcomes

Clinical trials often use time-to-event endpoints (e.g., overall survival), usually analyzed via the Log-Rank test or Cox proportional hazards models. These traditional methods require assumptions about proportional hazards or require waiting for a specific number of events. One can extend the randomization e-process to survival data, constructing a sequential Log-Rank test that updates wealth at every observed event.

Grünwald et al. (2021) developed a safe logrank test using e-values under a proportional hazards model with a prior on the hazard ratio. We attempted to construct a nonparametric approach where validity derives solely from randomization, not from a correctly specified hazard model. We call this time-to-event variant e-RTs.

8.1 Setup and Martingale Construction

Let N patients be randomized to treatment ($T = 1$) or control ($T = 0$). We observe outcomes over time. The "time" scale here is the distinct ordering of events. Let $t_1 < t_2 < \dots < t_k$ denote the times at which events occur.

At any event time t_j , we define the risk set $\mathcal{R}_j$ as the set of patients who have not yet had an event and have not been censored. Let $Y_1(t_j)$ and $Y_0(t_j)$ be the number of patients at risk in the treatment and control arms,

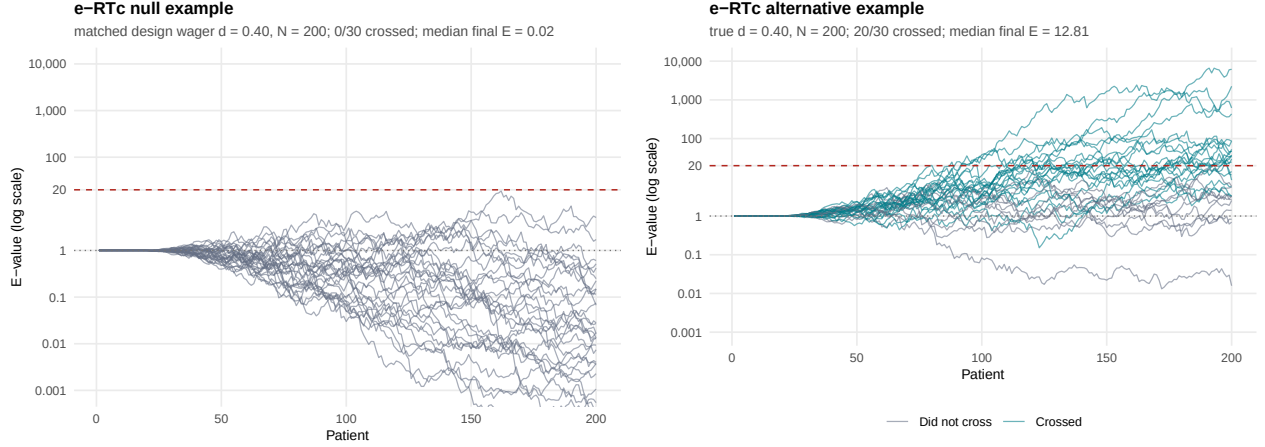


Figure 7: Trajectories of the continuous randomization e-process using the matched normal-shift design wager for a trial designed to detect a standardized mean difference of $d = 0.40$ with 80% power. Left: trajectories under the null hypothesis ($d = 0$), where wealth usually drifts below 1 and does not cross in the representative panel. Right: trajectories under the alternative hypothesis ($d = 0.40$), where wealth grows systematically and many paths cross the rejection threshold before the planned sample size is reached.

1 respectively.

2 Under the null hypothesis of no treatment effect, the probability that the event at time t_j comes from the
3 treatment arm, conditional on a failure occurring within $\mathcal{R}_j$, is simply the proportion of treated patients at risk:

$$p_j = \frac{Y_1(t_j)}{Y_1(t_j) + Y_0(t_j)}. \quad (42)$$

4 Let X_j be the indicator that the event at t_j is a treated patient ($X_j = 1$ if treated, 0 if control). Under the
5 null, X_j is a Bernoulli trial with probability p_j . We construct the martingale increment (score) as:

$$U_j = X_j - p_j. \quad (43)$$

6 Note that $\mathbb{E}[U_j | \mathcal{R}_j] = 0$. The corresponding one-step variance is

$$V_j = p_j(1 - p_j). \quad (44)$$

7 8.2 Betting Strategy

8 We wager on the sign and magnitude of U_j . If the treatment is beneficial, events will occur more slowly in the
9 treatment arm than expected under the null. Thus, the observed number of treatment events will be lower than
10 the expected number, leading to a negative trend in the cumulative sum of U_j .

11 For any predictable λ_j satisfying positivity of the multiplier, the wealth update at event j is:

$$W_j = W_{j-1} \times (1 + \lambda_j U_j). \quad (45)$$

12 Because $\mathbb{E}[U_j] = 0$, the expected multiplicative factor is 1 under the null. Thus, (W_j) is a test martingale. As
13 above, this validity engine is separate from the wager policy.

14 Let $Z_{j-1} = \sum_{k=1}^{j-1} U_k$ and $I_{j-1} = \sum_{k=1}^{j-1} V_k$. The original fixed-magnitude policy uses adaptive direction but
15 fixed size:

$$\lambda_j^{\text{fixed}} = c_j \lambda_{\max} \text{sign}(Z_{j-1}), \quad (46)$$

16 where $c_j \in [0, 1]$ is a ramping function similar to previous sections, and $\lambda_{\max} < 1$ is a cap on betting aggressiveness.

The adaptive policy estimates the log hazard ratio from the prior log-rank score:

$$\hat{\beta}_{j-1} = \frac{Z_{j-1}}{I_{j-1}}, \quad \hat{\theta}_{j-1} = \exp(\hat{\beta}_{j-1}), \quad (47)$$

and then converts $\hat{\theta}_{j-1}$ into a risk-set-specific wager. For a design or estimated hazard ratio θ , the event probability under a proportional hazards working alternative is:

$$q_j(\theta) = \frac{\theta p_j}{\theta p_j + (1 - p_j)}. \quad (48)$$

The GROW-style wager for the multiplier $1 + \lambda_j U_j$ is therefore:

$$\lambda_j^*(\theta) = \frac{q_j(\theta) - p_j}{p_j(1 - p_j)}. \quad (49)$$

The adaptive policy uses $c_j \kappa \lambda_j^*(\hat{\theta}_{j-1})$, with $\kappa = 1/2$ in the simulations below. The design/GROW-style policy instead prespecifies θ^* at the design stage and uses $c_j \lambda_j^*(\theta^*)$ throughout monitoring. This is directional: a design HR below 1 targets treatment benefit. As in the other e-RT variants, the working hazard ratio affects efficiency but not validity; under the null, U_j still has conditional mean zero for any predictable λ_j .

The parameters burn-in, ramp, $\lambda_{\max}$, and κ are tuning choices. Different choices will yield different operating characteristics. The validity of the test does not depend on these choices—only efficiency does.

8.3 Handling Staggered Entry

In clinical practice, patients are recruited over time (staggered entry), whereas the simplified simulations above generate survival times simultaneously. The log-rank score and the present betting strategy are naturally indexed by time since randomization, so a complete-follow-up simulation that analyzes $T_{\text{event}} - T_{\text{entry}}$ has the same event-time ordering as a simultaneous-entry simulation.

A dedicated check is provided in `R/simulations/erts_staggered_entry_check.R`. In 1,000 paired simulations with $N = 631$ and a true HR of 0.80, simultaneous entry and staggered entry analyzed on time since randomization were exactly identical: the maximum absolute difference in log final e-value was 0 and the first-crossing agreement was 100%. This exact agreement is expected in the complete-follow-up, no-censoring setting because analyzing time since randomization preserves the same event ordering and the same risk-set composition at each event; it is an algebraic equivalence in the paired simulation, not a claim about Monte Carlo noise vanishing in more general settings. The median final e-value was 23.0 and the ever-crossing probability was 66.0% in both analyses. A separate calendar-event-order stream, where risk sets contained only patients already enrolled at each calendar event, gave similar but not identical results (median final e-value 24.8; crossing probability 64.4%). Under HR = 1.00, the corresponding crossing rates were 2.6% for time-on-study order and 3.6% for calendar-event order. These checks support the simultaneous-entry simplification for the complete-follow-up, no-censoring simulations reported here, but they should not be over-interpreted as a full validation of delayed outcome availability, administrative censoring, or accrual-dependent interim decisions. Those settings require dedicated simulation.

8.4 Simulation Results

We simulated exponential survival trials with no censoring and 1:1 randomization. Event counts were chosen using the Schoenfeld log-rank event formula for 80% fixed-sample power at two-sided $\alpha = 0.05$. We compared the fixed-magnitude policy ($\lambda_{\max} = 0.25$), adaptive half-Kelly, and design/GROW-style wagers using underestimated, matched, and overestimated design hazard ratios. Results from 1,000 simulations per scenario are shown in Tables 10 and 11.

Type I error remained near or below nominal in these simulations. The largest null estimate was 5.4% for the fixed $\lambda_{\max} = 0.25$ policy in the HR = 0.90 planning scenario, which also has the largest event count (2,829) and therefore the most opportunities for multiplicative compounding; with 1,000 simulations, this is within Monte

Table 10: e-RTs Type I error simulations under $HR = 1$. Planning event counts use the Schoenfeld log-rank event formula for 80% fixed-sample power at two-sided $\alpha = 0.05$.

Planning HR	Policy	Events	Type I error	Median final e-value
0.70	Fixed 0.25	247	0.6%	0.23
0.70	Adaptive half	247	0.1%	0.73
0.70	Design matched	247	2.2%	0.06
0.80	Fixed 0.25	631	3.5%	0.01
0.80	Adaptive half	631	0.5%	0.63
0.80	Design matched	631	2.3%	0.03
0.90	Fixed 0.25	2,829	5.4%	0.00
0.90	Adaptive half	2,829	1.2%	0.51
0.90	Design matched	2,829	4.3%	0.02

Table 11: e-RTs power and crossing diagnostics. Sample sizes use the Schoenfeld log-rank event formula for 80% fixed-sample power at two-sided $\alpha = 0.05$. Type M and Type S are summarized among trials that crossed the e-value threshold. Type M is computed on the $|\log(HR)|$ scale.

True HR	Policy	Wager HR	Power	Crossing	HR at crossing	Type M	Type S
0.70	Fixed 0.25	–	46.8%	186	0.62	1.35	0.0%
0.70	Adaptive half	–	28.2%	170	0.57	1.56	0.0%
0.70	Design under	0.85	17.5%	199	0.57	1.57	0.0%
0.70	Design matched	0.70	62.7%	158	0.64	1.27	0.0%
0.70	Design over	0.55	61.7%	124	0.63	1.30	0.0%
0.80	Fixed 0.25	–	61.2%	311	0.73	1.42	0.0%
0.80	Adaptive half	–	38.2%	380	0.70	1.57	0.0%
0.80	Design under	0.90	40.2%	472	0.73	1.40	0.0%
0.80	Design matched	0.80	70.8%	328	0.75	1.32	0.0%
0.80	Design over	0.70	66.3%	249	0.73	1.43	0.0%
0.90	Fixed 0.25	–	37.3%	647	0.80	2.08	0.5%
0.90	Adaptive half	–	41.6%	1,684	0.85	1.56	0.2%
0.90	Design under	0.95	53.8%	2,120	0.87	1.33	0.0%
0.90	Design matched	0.90	75.4%	1,366	0.87	1.30	0.0%
0.90	Design over	0.85	70.4%	1,014	0.85	1.49	0.0%

Carlo uncertainty around the 5% target. The matched design wager was the most powerful policy at the fixed-sample event counts: 75.4% power for $HR = 0.90$, 70.8% for $HR = 0.80$, and 62.7% for $HR = 0.70$. These are below the nominal 80% fixed-sample log-rank design because the e-process must cross the anytime-valid threshold before or by the maximum event count. The fixed $\lambda_{\max} = 0.25$ policy remained competitive, especially at $HR = 0.80$, but it was less efficient than a matched design wager. Adaptive half-Kelly was conservative, reflecting the difficulty of estimating the hazard-ratio scale early from sparse events.

Misspecification behaved as expected. Underestimating the treatment effect delayed crossings and reduced power. Overestimating the effect crossed earlier and sometimes retained high power, but with greater Type M exaggeration. Type S error was essentially absent: only the two-sided fixed and adaptive policies at $HR = 0.90$ showed small wrong-direction crossing rates below 1%. Examples for the fixed-magnitude policy at $HR = 0.80$ are shown in Figure 10.

Figure 8 visualizes the same power comparison. The fixed $\lambda_{\max} = 0.25$ policy remains a useful baseline, while the matched design wager comes closest to the conventional 80% log-rank target across the tested hazard ratios.

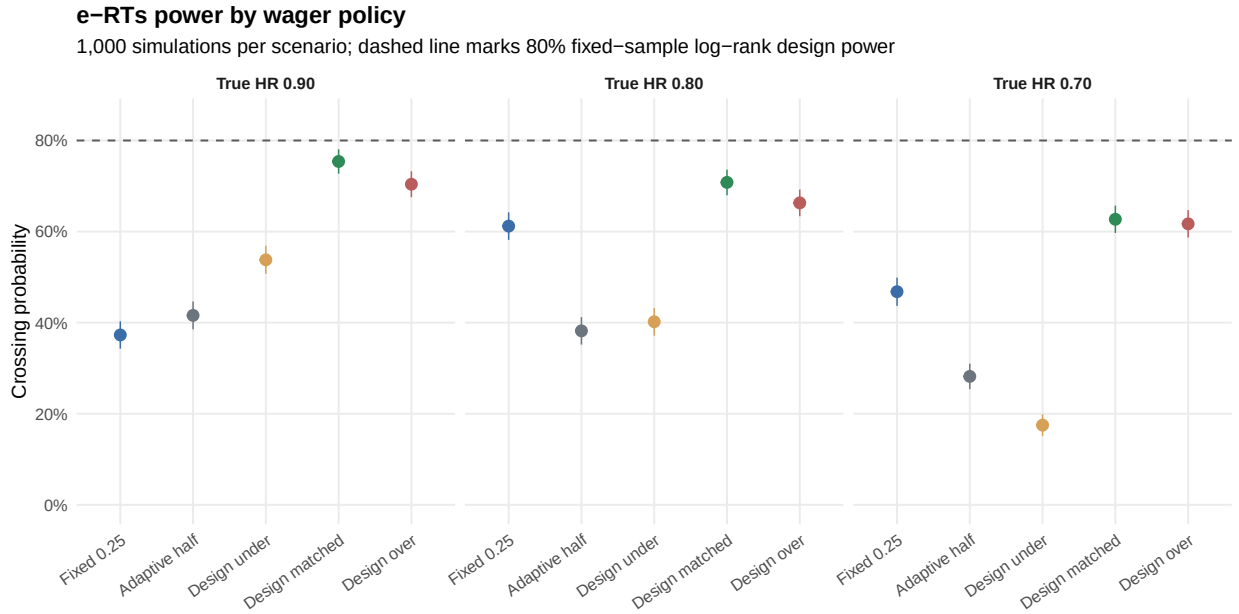


Figure 8: Power of e-RTs wager policies across hazard-ratio scenarios. Points show crossing probability from 1,000 simulations per scenario, with approximate 95% Monte Carlo intervals. The dashed line marks the 80% power target used for the fixed-sample log-rank event calculation.

Figure 9 shows the corresponding Type M error at crossing. Even when Type I error is controlled, the apparent hazard ratio at crossing can be exaggerated, especially when the true effect is small or the wager is aggressive.

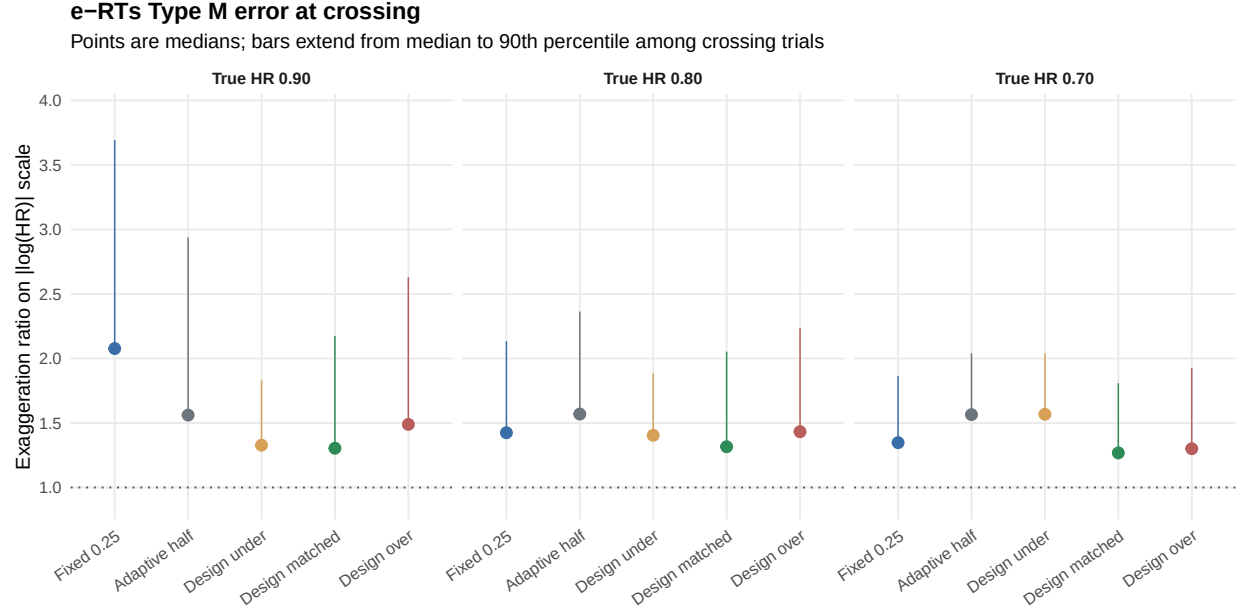


Figure 9: Type M error at first e-RTs threshold crossing. Points show the median exaggeration ratio on the $|\log(\text{HR})|$ scale among crossing trials; vertical bars extend from the median to the 90th percentile. The dotted line marks no exaggeration.

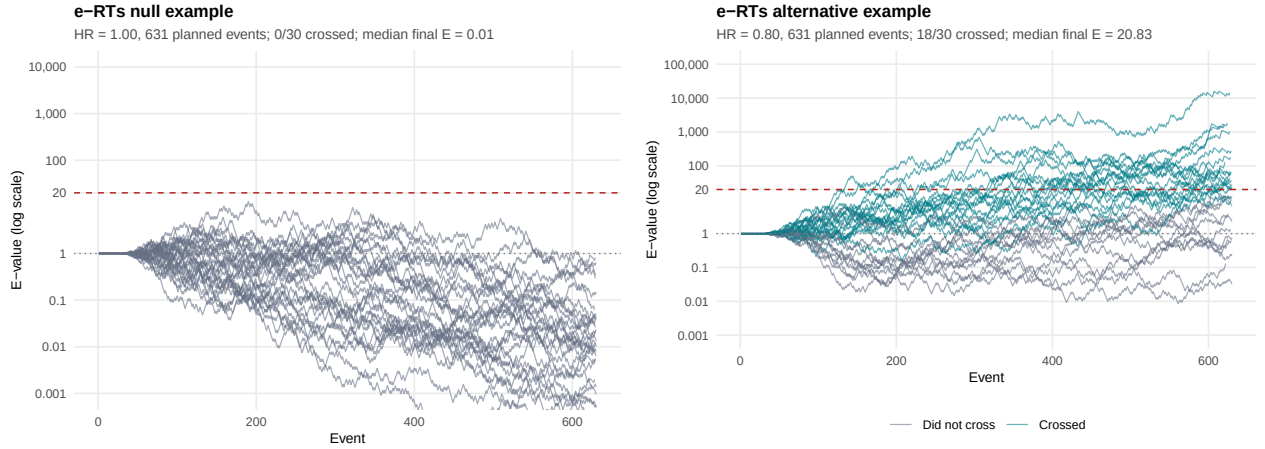


Figure 10: Trajectories of the fixed-magnitude e-RTs process for a trial designed to detect a Hazard Ratio of 0.80 with 80% fixed-sample log-rank power (631 events). Left: trajectories under the null hypothesis ($\text{HR} = 1.00$), where wealth fluctuates randomly. Right: trajectories under the alternative hypothesis ($\text{HR} = 0.80$), where many paths grow and cross the rejection threshold. The red dashed line represents the rejection threshold ($1/\alpha = 20$).

9 Betting Strategy Design Across Endpoints

The active e-RT variants use different betting strategies. This is not a validity distinction: all variants rely on the same predictable-martingale argument. The difference is an efficiency distinction. A wager that is reasonable for one endpoint may be inefficient for another because the wealth product is updated on a different scale.

9.1 Update Density and Over-Betting

The wealth process is a product of multipliers. When a wager is too aggressive for the true effect, unfavorable multipliers are compounded repeatedly. The practical cost depends on update density: survival and event-only methods update only when events occur, whereas binary and continuous endpoints update for every patient. This does not change Type I error control, but it can strongly affect power and the distribution of final e-values.

9.2 Sparse-Update Endpoints: e-RTs and e-RTe

For e-RTs, wealth updates at observed failures. This makes fixed or design-calibrated wagers comparatively tolerable, because the number of multiplications is the number of events rather than the number of randomized patients. The e-RTs simulations in Table 11 support this interpretation. The fixed $\lambda_{\max} = 0.25$ policy is not uniformly best, but it remains a useful baseline: at HR = 0.80, it crossed in 61.2% of simulations. The matched design wager was more efficient when the design HR was credible, crossing in 70.8% of simulations at HR = 0.80 and 75.4% at HR = 0.90. Adaptive half-Kelly was conservative in these simulations, reflecting the difficulty of estimating a hazard-ratio scale early from risk-set data.

The event-only e-RTe also updates sparsely, but on a different scale: the update stream is the sequence of events rather than risk-set failures. This permits a more aggressive default than e-RTb, and the event-coin parameter is naturally bounded between 0 and 1. The tuning simulations in Table 4 show, however, that sparse updating creates its own design problem. If the burn-in and ramp consume much of the planned event stream, e-RTe may not begin meaningful betting soon enough. Thus e-RTe can use full-Kelly intensity, but its burn-in and ramp should be chosen with the expected number of events in mind.

9.3 Dense-Update Endpoints: e-RTb and e-RTc

For e-RTb, wealth updates for every randomized patient. This makes naive over-betting costly. Table 12 isolates this issue using fixed deviations of the assignment-prediction wager from 0.5 under a true 5pp ARR: for events the wager is $0.5 - m$ and for non-events it is $0.5 + m$. Here m is a betting magnitude, not an ARR. A 0.05 deviation crossed in 57.3% of simulations, whereas twofold and threefold larger deviations crossed in only 24.3% and 13.2% of simulations, with median final e-values effectively zero. This stress test is intentionally narrow: it does not say that all fixed wagers are poor. Rather, it shows why the adaptive binary default uses half-Kelly shrinkage and why design-fixed binary wagers should be calibrated to clinically plausible effects.

For e-RTc, the update density is also every patient, and the wager additionally varies with an observation-level score. The adaptive sign-direction rule therefore remains conservative, especially for small effects. Table 9 shows that at true $d = 0.20$, adaptive e-RTc crossed in only 9.8% of simulations and had substantial Type M inflation among crossings. A matched normal-shift design wager improved power to 73.4% and reduced median Type M to 1.28, but this efficiency depends on the design model being credible. Continuous endpoints therefore emphasize the same separation seen elsewhere: validity is supplied by randomization and predictability; efficiency depends on the wager.

9.4 Design Principle

Table 13 summarizes the practical hierarchy across active variants. Sparse-update methods tolerate more aggressive wagers because there are fewer opportunities for over-betting to compound. Dense-update methods require more shrinkage or stronger design justification. This is a statement about power and stability, not validity: any predictable wager remains valid under the null, but poorly calibrated wagers may spend their wealth in the wrong places.

The stress table below gives the binary example underlying this principle.

The summary table below should be read as design guidance rather than a theorem.

Table 12: Binary: cost of naive over-betting (true ARR = 5%, $N = 2,942$). Fixed wager magnitudes are deviations from 0.5 in the assignment-prediction wager, not ARR design effects. Rows summarize 1,000 fixed-seed simulated trials.

Fixed wager magnitude	Scale vs 0.05 wager	Power	Median Final E-value
0.05	1×	57.3%	1.31
0.10	2×	24.3%	≈ 0
0.15	3×	13.2%	≈ 0
0.20	4×	11.6%	≈ 0

Table 13: Summary: wager-policy design considerations by e-RT variant.

Property	e-RTs	e-RTe	e-RTb	e-RTc
Update unit	Failure	Event	Patient	Patient
Default policy	Fixed 0.25	Full Kelly	Half Kelly	Sign direction
Design-wager role	HR planning	Event-rate planning	ARR planning	Normal-shift model
Over-betting sensitivity	Lower	Moderate	High	High
Main tuning issue	HR misspecification	Event-stream length	Kelly shrinkage	Scale/model choice

10 Discussion

The e-RT family comprises nonparametric sequential tests for randomized trials based on the betting framework for e-values (i-bet Duan et al. (2022)). All active variants require only that treatment assignment is randomized—no distributional assumptions about outcomes are needed for validity. This makes them robust complements to model-based analyses. This manuscript focuses on binary outcomes (e-RTb), event-only monitoring (e-RTe), continuous endpoints (e-RTc), and time-to-event analyses (e-RTs).

10.1 Operating characteristics

Across the simulation-validated variants, simulations demonstrate proper Type I error control, confirming the theoretical guarantee from the martingale property. Power varies by variant and scenario: for binary outcomes, approximately 50% power for early stopping in trials designed with 80% frequentist power, and 63–66% in trials designed with 90% power. The event-only variant (e-RTe) trades information for operational simplicity because non-events are not used by the e-process. The e-RTc simulations show a wager-policy tradeoff on the Cohen’s d scale: a matched parametric design wager can be much more powerful than the agnostic adaptive wager, while misspecification changes both power and Type M error at crossing. The e-RTs simulations extend this pattern to time-to-event monitoring: a design hazard-ratio wager is most efficient when calibrated, while the fixed-magnitude policy remains a robust baseline and adaptive half-Kelly is conservative.

When early stopping occurs across all variants, it typically happens at approximately 45–56% of the planned sample size or event count.

These results should be interpreted carefully. From a clinical trial perspective, it is uncertain whether these methods can replace traditional frequentist or Bayesian paradigms, but they may provide a continuous monitoring option that requires no alpha spending and no prespecified interim analysis schedule. If the e-process crosses its threshold, one could consider stopping the trial. If it does not cross, the trial may proceed to its planned conclusion and primary analysis.

10.2 Traditional statistics at crossing

When an e-RT crosses its threshold ($\geq 1/\alpha$), one may ask what a traditional frequentist analysis would report at that exact moment. In the simulated crossing trials, conventional analyses usually moved in the same direction and often produced small p-values or confidence intervals excluding the null. This is reassuring, but it should be treated as descriptive rather than inferential.

1 However, there is a critical distinction. The traditional analysis at crossing is *formally invalid* because the
2 analyst observed the data at multiple time points and stopped at a favorable one. The frequentist p-value is
3 anti-conservative (too small) and the standard confidence interval has less than 95% coverage at this data-driven
4 stopping time. The e-value, by contrast, is valid regardless of the stopping rule—this is the entire point of the
5 anytime-valid guarantee. The e-value itself carries the inferential weight; traditional statistics at crossing are best
6 used as clinical descriptors and diagnostic checks.

7 10.3 What is the null hypothesis being tested?

8 This approach tests whether treatment assignment can be predicted from outcomes—equivalently, whether out-
9 comes are exchangeable between arms. Under the null, $Y_i \perp T_i$ at each observation: knowing the outcome provides
10 no information about which arm the patient belongs to. This is neither Fisher’s sharp null (every individual has
11 exactly zero treatment effect) nor the weak null of equal population means.

12 A useful analogy is a casino. Under the null, the house is fair: no betting strategy can systematically grow
13 wealth. The e-value quantifies accumulated evidence that the game is beatable. Rejecting the null means we have
14 found a profitable strategy—outcomes predict assignments better than chance. The user is also referenced to the
15 pioneer lessons by Ramdas (2021).

16 This framing clarifies both the method’s strength and its limitation. The strength is generality: any departure
17 from exchangeability—constant effects, heterogeneous effects, or even randomization failures—makes outcomes
18 informative and wealth grows. The limitation is that the test detects only departures that a *cumulative backward-*
19 *looking* betting strategy can exploit. In practical terms, the null hypothesis is best understood as: “there exists
20 no stable, learnable relationship between outcomes and treatment assignment that persists long enough for an
21 adaptive strategy to exploit.” This is appropriate for most trials where effects are expected to be consistent over
22 enrollment, but investigators should be aware that non-stationary effects and abrupt direction reversals are blind
23 spots for the current cumulative wager; recency-weighted extensions are discussed below.

24 10.4 Relationship to existing work

25 The betting framework for hypothesis testing was developed by Shafer (2021). E-values and e-processes have
26 been extensively studied (Vovk and Wang, 2021; Ramdas et al., 2022; Ramdas and Wang, 2025). Duan et al.
27 (2022) introduced interactive rank testing by betting (i-bet), which applies the betting framework directly to
28 randomized experiments: an analyst sequentially bets on treatment assignments based on observed outcomes,
29 with wealth forming a test martingale under the null. The binary approach implements this framework with a
30 specific adaptive betting strategy tied to outcome values. Betting approaches have been established for estimating
31 means of bounded random variables (Waudby-Smith and Ramdas, 2023). The continuous extension adapts these
32 principles to the two-sample randomization setting using a standardization strategy.

33 Sokolova and Sokolov (2026a) and the accompanying `evalinger` implementation (Sokolova and Sokolov,
34 2026b) are especially important comparators for the present work. The first draft of e-RT was dated December
35 4, 2025, so the randomization-betting construction developed independently of that manuscript. This chronol-
36 ogy is noted only to clarify the origin of e-RT, not to diminish the importance of their contribution. Their
37 work provides a mature design-calibrated perspective on clinical-trial e-processes, particularly through growth-
38 rate-optimal (GROW) wagers chosen from prespecified alternatives. The present work uses their contribution
39 as a central point of comparison: e-RT is effect-size agnostic by default, whereas GROW-style design wagers
40 encode a planned alternative to improve expected evidence growth. The two views are compatible because ran-
41 domization supplies the validity engine and the wager policy supplies the efficiency profile. In this manuscript,
42 design-calibrated wagers are therefore presented as optional efficiency tools within e-RT, not as replacements for
43 the adaptive agnostic monitor.

44 Koning (2025) develops e-values for group invariance, including permutation tests, using batch-based likelihood
45 ratio statistics normalized by permutation expectations. Grünwald et al. (2021) developed the ‘Safe Log-rank
46 Test’ based on evaluating likelihood ratios with specific priors on the hazard ratio to ensure growth rate optimality.
47 In contrast, e-RTs constructs a linear test martingale directly from the log-rank score increment. Rather than
48 requiring likelihood integration or a correct proportional-hazards model for validity, e-RTs derives validity from

the randomization probabilities within each risk set. The wager policy may be fixed, adaptive from the prior score, or design-calibrated from a prespecified hazard ratio; the working hazard ratio affects efficiency, not Type I error control.

Pairwise and prioritized composite endpoints remain important future work. Generalized pairwise comparisons and win-ratio methods compare treatment and control patients using clinically prioritized rules (Buyse, 2010; Wang and Pocock, 2016). A first exploratory implementation considered disjoint treatment-control pair signs as an e-process, with `BuyseTest` used as an all-pairs GPC reference (Ozenne and Peron, 2025). We no longer include that prototype as an active e-RT variant because it bets on pairwise outcome direction after treatment assignment is known, rather than betting on randomized assignment. A future e-RT-compatible pairwise method should keep the randomization-test structure: form pairs predictably, observe the pair outcomes, determine which member has the preferable clinical outcome, and then bet whether that member was randomized to treatment. The former prototype is retained in the repository as an exploration, not as support for the active manuscript claims.

The e-RTb shares the same martingale foundation as i-bet but differs in key respects: it operates prospectively as patients enroll rather than retrospectively on completed data; it requires no covariates or working models; and it uses betting fractions that adapt continuously to running outcome estimates rather than fixed-magnitude wagers guided by covariate-based predictions. This yields a simpler method that may be suitable for real-time trial monitoring.

It is possible that some of the concepts here were discussed by other authors in different contexts that were not immediately available for this author. Reader is encouraged to reach out if that is the case, and the author will happily adjust accordingly.

10.5 Limitations

Several limitations should be noted. This is an experimental method under development. Simulations are not exhaustive, and the operating characteristics reported are specific to the scenarios tested. It is uncertain how these methods would behave in more complex models, including competing risk and multi-state settings, which are deliberately deferred from the active scope.

The methods test only whether there are differences between arms; they do not provide point estimates or confidence intervals. The adaptive learning of $\hat{\delta}$ requires a burn-in period during which little evidence accumulates. For trials where parametric assumptions are plausible, model-based sequential methods will generally have better power. Our simulations used specific betting strategies; other choices may yield different operating characteristics. The betting strategy design section provides guidance on strategy selection, but optimal calibration for specific clinical scenarios remains an open question. Finally, it is unclear how these methods will behave when heterogeneity in treatment effects exists or there are temporal instabilities in effect size. Extensions to relative effect size approaches (e.g., odds ratio) are under development.

10.6 Future Directions

Several extensions merit exploration. First, the current betting strategy uses a cumulative estimate $\hat{\delta}$ that weights all historical observations equally. This makes the method vulnerable to time-varying effects: if treatment benefit reverses to harm mid-trial, the strategy continues betting on stale information and wealth erodes despite continuous violation of exchangeability. Adaptive weighting schemes—such as exponential decay, rolling windows, or hybrid estimators blending long-term and recent signals—could improve robustness to non-stationary effects. Second, the betting intensity could itself adapt to recent performance: increasing λ during sustained wealth growth (exploiting a confirmed edge) and dampening it following drawdowns (protecting against regime change). Third, pairwise and prioritized composite endpoints may be compatible with e-RT if formulated as assignment-prediction randomization tests: form pairs predictably, observe pair outcomes, determine the clinically preferable member, and then bet on whether that member was randomized to treatment. Fourth, the multi-state and universal-signal ideas from earlier drafts remain useful but need a separate treatment, especially if transitions or patient-level repeated contributions create dependence concerns beyond the simple binary-signal story. These

1 refinements trade power under stable effects against robustness to drift and complexity, and the optimal balance
2 likely depends on the clinical context.

3 **10.7 Conclusion**

4 E-processes provide anytime-valid sequential inference for randomized trials using only the guarantee of random-
5 ization. The e-RT family separates this validity engine from the wager policy used for efficiency. Its default
6 contribution is effect-size-agnostic monitoring: a trial can be monitored continuously without specifying the
7 treatment effect the e-process is trying to exploit. Design-calibrated wagers, including GROW-style wagers, can
8 improve power when the design alternative is credible, but they are optional efficiency tools rather than validity
9 requirements. This makes e-RT a conservative, transparent complement to traditional trial monitoring rather
10 than a replacement for the final model-based analysis.

11 **11 Disclaimers and Version Control**

12 **11.1 Disclaimer**

13 This is an experimental method under development. Application to real patients should only be considered under
14 surveillance from an experienced statistician and remain strongly discouraged at this point by the author. The
15 author is not responsible for consequences of use of this method.

16 **11.2 LLM use statement**

17 Large language models were extensively used in this work. Before using LLMs, the author formulated the
18 central idea that e-value and e-process machinery could be used to bet on randomized assignments from observed
19 outcomes, yielding a randomization-based continuous monitoring tool. The author then uploaded the references
20 in this manuscript to Gemini 3.0 Pro for brainstorming and drafting support. Subsequent versions were refined,
21 tested, and debugged using Claude 4.5 Opus and ChatGPT 5.1 Pro. Gemini 3.0 Pro aided with coding for survival
22 approach. Claude Opus 4.6 aided with the deaths-only extension and the wager asymmetry analysis in V6, and
23 with renaming, generalization of e-RTd to e-RTe, and the e-RTu universal abstraction in V7. OpenAI Codex
24 aided the Version 8 repository organization, simulation refactoring, wager-policy comparisons, pairwise endpoint
25 explorations, e-RTc and e-RTs design-wager implementation, and manuscript cleanup.

26 **11.3 Acknowledgments**

27 The author is thankful to Aaditya Ramdas for their thoughtful comments on the first version and for pointing
28 out previous literature to the author.

29 **11.4 Code and Data Availability**

30 The manuscript source, R implementation files, simulation scripts, generated CSV result tables, manuscript tables,
31 and figures are maintained in the project repository: <https://github.com/fzampier/erandtest>. This reposi-
32 tory is the source of truth for the computational appendix, rather than duplicating code inside the manuscript.

33 **11.5 Version Control**

- 34 1. First Version (Dec 04, 2025)
- 35 2. Second Version (Dec 07, 2025): Minor text adjustments; removed claim on sharp null.
- 36 3. Third Version (Dec 11, 2025): Added continuous and survival endpoints; text adjustments.
- 37 4. Fourth Version (Dec 17, 2025): Correction on signal direction on e-RTC. Text adjustments.
- 38 5. Fifth Version (Dec 31, 2025): Added multi-state extension (e-RTms).

6. Sixth Version (Feb 15, 2026): Added deaths-only monitoring (e-RTd). Added betting strategy design section explaining wager asymmetry across variants. Added traditional statistics at crossing discussion. Updated abstract and introduction to cover all five variants. Reordered sections.
7. Seventh Version (Mar 08, 2026): Renamed binary e-RT to e-RTb after its introduction as the prototype. Generalized e-RTd (deaths-only) to e-RTe (event-only), broadening applicability beyond mortality. Added e-RTu (universal) section describing a domain-agnostic betting engine abstraction (under development). Updated all cross-references and discussion to reflect six variants.
8. Eighth Version (in preparation): Separated randomization validity from wager policy; added adaptive, design-fixed, misspecified-design, and oracle wager simulations for e-RTb/e-RTe; added Type M error at crossing diagnostics; added parametric normal-shift design wagers for e-RTc; added fixed, adaptive, and design-calibrated wager-policy simulations for e-RTs with Type M and Type S diagnostics; committed simulation result tables for reproducibility; deferred e-RTms, e-RTu, and pairwise-comparison prototypes from the active manuscript scope; and removed the embedded code appendix in favor of the project repository.

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